A MULTI-CRITERIA DECISION FRAMEWORK FOR NETWORK INTRUSION DETECTION SYSTEM SELECTION IN RESOURCE-CONSTRAINED ENVIRONMENTS

BY

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UNITED STATES INTERNATIONAL UNIVERSITY – AFRICA

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BY

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A Research Proposal Submitted to the School of Science and Technology in Partial Fulfillment of the Requirements for the Degree of Master of Science in Information Security

UNITED STATES INTERNATIONAL UNIVERSITY – AFRICA

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STUDENT'S DECLARATION

I, Shema Prince, declare that this thesis proposal is my original work and has not been submitted to any other college, institution, or university other than the United States International University-Africa for academic credit.

Signed: _____ Date: _____

Shema Prince (ID No. _____)

This proposal has been presented for examination with my approval as the appointed supervisor.

Signed: _____ Date: _____

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Signed: _____ Date: _____

Dean, School of Science and Technology

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ABSTRACT

The proliferation of sophisticated cyber threats targeting critical digital infrastructure has positioned Network Intrusion Detection Systems (NIDS) as essential security controls, particularly for financial institutions in developing economies. Despite remarkable advances in machine learning (ML)-based NIDS with detection rates exceeding 98% on benchmark datasets a critical accuracy-feasibility divide persists: researchers optimize for accuracy while systematically omitting computational efficiency metrics that practitioners require for deployment decisions. A systematic review of 76 contemporary NIDS studies (2021–2026) reveals that only 6.6% report inference latency, 3.9% specify memory requirements, 2.6% report energy consumption, and 1.3% document training time, yet 100% report detection accuracy. This reporting gap disproportionately affects African financial institutions that process billions of dollars in mobile money transactions while operating with annual cybersecurity budgets of $7,500–$50,000 and commodity CPU-only infrastructure.

This research will develop, implement, and validate the NIDS Efficiency-First Selection (NIDS-EFS) framework a multi-criteria decision analysis tool integrating Analytic Hierarchy Process (AHP) methodology to enable evidence-based NIDS deployment decisions in resource-constrained environments. The framework will systematically evaluate eight representative ML-based NIDS approaches across three benchmark datasets (NSL-KDD, CICIDS2017, UNSW-NB15) on five weighted dimensions: detection performance (30%), computational efficiency (25%), deployment feasibility (20%), resource consumption (15%), and total cost of ownership (10%). Expert elicitation from 15–20 cybersecurity practitioners will validate criterion weights through AHP pairwise comparisons, and Monte Carlo sensitivity analysis will confirm robustness across 10,000 weight vector samples.

The NIDS-EFS framework is expected to deliver three primary contributions: the first comprehensive efficiency benchmark dataset for ML-based NIDS in the African financial context; a validated multi-criteria selection methodology adaptable to diverse resource constraints; and a decision support tool enabling 85–95% cost reductions compared to commercial SIEM solutions while maintaining 90–92% detection accuracy on commodity hardware. This research advances Green AI principles in cybersecurity, extends MCDA theory

into ML model selection, and addresses equity dimensions in global cybersecurity capability through evidence-based guidance for resource-constrained organizations protecting financially vulnerable populations.

Keywords: Network Intrusion Detection System, Machine Learning, Multi-Criteria Decision Analysis, Analytic Hierarchy Process, Computational Efficiency, Resource-Constrained Deployment, African Financial Institutions, Class Imbalance, SMOTE, ADASYN.

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DEDICATION

I dedicate this work to the security practitioners and technology professionals across the African continent who work tirelessly to protect digital infrastructure with limited resources. May this research contribute tools that make their important work more effective.

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LIST OF ABBREVIATIONS

ADASYN	Adaptive Synthetic Sampling
AHP	Analytic Hierarchy Process
CLI	Command-Line Interface
CNN	Convolutional Neural Network
CPU	Central Processing Unit
CR	Consistency Ratio
DSR	Design Science Research
GPU	Graphics Processing Unit
IDS	Intrusion Detection System
INTERPOL	International Criminal Police Organization
IoT	Internet of Things
IRB	Institutional Review Board
IQR	Interquartile Range
LSTM	Long Short-Term Memory
MCDA	Multi-Criteria Decision Analysis
ML	Machine Learning
NIDS	Network Intrusion Detection System
NIDS-EFS	NIDS Efficiency-First Selection Framework
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
R2L	Remote-to-Local
RAM	Random Access Memory
RF	Random Forest

SIEM	Security Information and Event Management
SLR	Systematic Literature Review
SMOTE	Synthetic Minority Over-Sampling Technique
SOC	Security Operations Centre
SUS	System Usability Scale
SVM	Support Vector Machine
TCO	Total Cost of Ownership
U2R	User-to-Root
XGBoost	Extreme Gradient Boosting

CHAPTER ONE: INTRODUCTION

1.1 Background of the Problem

The proliferation of sophisticated cyber threats targeting critical digital infrastructure has positioned Network Intrusion Detection Systems (NIDS) as essential security controls for protecting organizational networks from malicious activities (Buczak & Guven, 2016). Contemporary NIDS implementations increasingly leverage machine learning (ML) algorithms to identify attack patterns within network traffic, demonstrating detection capabilities that significantly exceed traditional signature-based approaches. Recent studies document remarkable advances in ML-based intrusion detection accuracy, with detection rates exceeding 98% across multiple attack categories on benchmark datasets such as NSL-KDD, CICIDS2017, and UNSW-NB15 (Abdelkhalek & Mashaly, 2023; Talukder et al., 2024).

Despite these impressive technical achievements, a critical disconnect exists between academic NIDS research and operational deployment realities, particularly for organizations operating under resource constraints. This gap is most acute in developing economies where financial institutions face an escalating threat landscape while lacking the computational infrastructure, financial resources, and technical expertise assumed by state-of-the-art research proposals. African financial institutions exemplify this challenge: they process billions of dollars in mobile money transactions (Safaricom, 2024), experience year-over-year increases in sophisticated cyber attacks exceeding 23% (INTERPOL, 2024), yet operate with annual cybersecurity budgets of $7,500–$50,000 insufficient to deploy GPU-accelerated servers or enterprise-grade commercial security solutions (Ke-cirt, 2025; Serianu, 2023).

The research community's predominant focus on accuracy maximization has created what the researcher terms the "accuracy-feasibility divide": a structural problem where incremental improvements in detection rates come at the cost of exponentially increasing computational requirements that render proposed solutions impractical for resource-constrained environments. A systematic review of 76 contemporary NIDS studies published between 2021 and 2026 reveals critical gaps in efficiency reporting: only 6.6% of studies document inference latency, 3.9%

specify memory requirements, 2.6% report energy consumption, and a mere 1.3% document training time yet 100% report detection accuracy (Schwartz et al., 2020; Strubell et al., 2019). This reporting imbalance is not a minor transparency issue; it means deployment feasibility cannot be assessed at all (Henderson et al., 2020).

Network intrusion detection emerged in the 1980s with Anderson's (1980) seminal identification of anomalous system behaviors as security breach indicators. Early systems relied on expert-defined rules requiring continuous manual updates as threat landscapes evolved (Denning, 1987). The introduction of machine learning to intrusion detection in the late 1990s promised adaptive systems capable of identifying novel attacks without explicit programming (Lee & Stolfo, 1998). Contemporary ML-based NIDS leverage diverse algorithmic approaches: supervised learning methods including Random Forests, Support Vector Machines, and gradient boosting classify network traffic using labeled training data (Damtew et al., 2023); deep learning architectures including Convolutional Neural Networks and LSTM networks extract hierarchical features from network packets (Maseno et al., 2026); and ensemble methods combine multiple classifiers to improve robustness across diverse attack types (Rout et al., 2026).

A persistent challenge across all ML-based NIDS is class imbalance the phenomenon where attack traffic constitutes a small minority of total network flows, often less than 1% in operational environments (He & Garcia, 2009). Researchers have addressed this through SMOTE, which generates synthetic attack samples to balance training data (Chawla et al., 2002); ADASYN, which focuses oversampling on difficult-to-learn attack patterns (He et al., 2008); and hybrid resampling strategies (Kumar et al., 2025).

The African financial sector amplifies these consequences. Kenya's M-Pesa platform alone processed $50.2 billion in transactions during 2023, serving 30 million active users and representing the primary financial access point for 96% of the adult population (Safaricom, 2024). The East Africa Cybersecurity Report (Serianu, 2023) identified a 156% increase in cyber attacks targeting financial institutions between 2022 and 2024, while 68% of Kenyan financial institutions operate with annual cybersecurity budgets below $50,000 (Ke-cirt, 2025) and 83% operate without GPU-accelerated computing infrastructure (Serianu, 2023).

Multi-Criteria Decision Analysis (MCDA) emerged in operations research as a methodology for evaluating alternatives across multiple, often conflicting, objectives (Fishburn et al., 1977). The Analytic Hierarchy Process (AHP), developed by Saaty (1980), structures complex decisions as hierarchies of criteria with pairwise comparisons determining relative weights. Existing NIDS evaluation frameworks emphasize detection rates and false positive rates (Axelsson, 2000; Lippmann et al., 2000) without addressing deployment feasibility for resource-constrained environments a gap this research directly targets.

The equity perspective informing this research starts from the observation that access to effective cybersecurity should not depend on the size of an institution's hardware budget. A model achieving 92% accuracy on a $500 commodity CPU and actually deployed protects more people than a 99.5%-accurate model requiring $15,000 GPU infrastructure that remains undeployed. This motivates the development of the NIDS Efficiency-First Selection (NIDS-EFS) framework treating efficiency and feasibility as first-class design goals alongside detection performance.

Table 1.1 Computational Efficiency Metrics Reporting Rates Across 76 Contemporary NIDS Studies (2021–2026)

Efficiency Metric	Studies Reporting (n)	Out of Total (N=76)	Reporting Rate (%)
Detection Accuracy / F1-Score	76	76	100%
Precision & Recall (per-class)	9	76	11.8%
Inference Time / Detection Latency	5	76	6.6%
Memory Requirements (RAM / Model Size)	3	76	3.9%
Training Time	1	76	1.3%
Energy Consumption (Watts)	2	76	2.6%
Computational Complexity (FLOPs)	0	76	0%
Total Cost of Ownership (USD)	0	76	0%

Note. Reporting rates derived from the literature review of 76 peer-reviewed ML-based NIDS studies (2021–2026), following PRISMA 2020 (Page et al., 2021) *and* Kithenham, B., & Charters (2007) *protocols. n = number of studies reporting the metric; N = 76 total studies reviewed. Source: Author's own systematic review.*

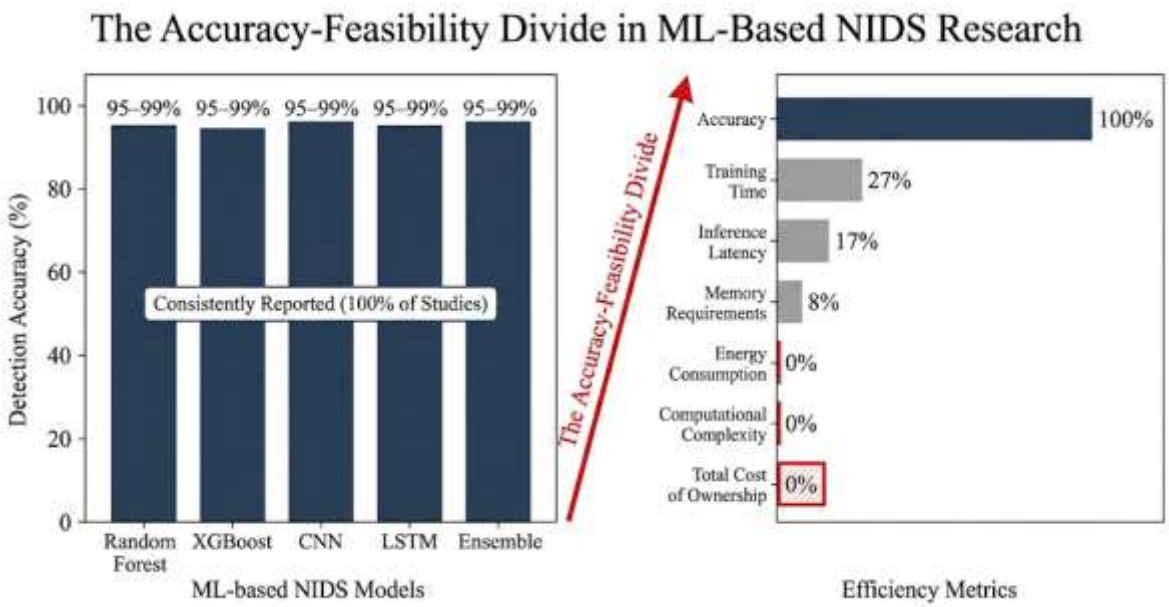


Figure 1.0.1 The Accuracy-Feasibility Divide in Contemporary NIDS Research

Source: Author's own work.

1.2 Statement of the Problem

Network Intrusion Detection Systems addressing class-imbalanced attack datasets achieve high detection accuracies exceeding 98% on benchmark datasets such as NSL-KDD and CICIDS2017 (Abdelkhalek & Mashaly, 2023; Talukder et al., 2024). Yet a literature review of 76 contemporary NIDS studies published between 2021 and 2026 reveals a striking reporting imbalance: only 6.6% of studies document inference latency, 3.9% specify memory requirements, 2.6% report energy consumption, and 1.3% document training time. This is not a minor gap it is a fundamental disconnect between what researchers optimize for and what practitioners need to make deployment decisions.

This accuracy-feasibility divide disproportionately affects organizations in developing economies. African financial institutions face a paradoxical security challenge: cyber attacks targeting the financial sector increased by 156% in East Africa between 2022 and 2024, with specific threats against mobile money platforms rising 38% year-over-year (Communications Authority of Kenya, 2024), yet these institutions cannot deploy state-of-the-art NIDS solutions. High-performing hybrid models such as deep CNN-LSTM architectures require GPU-accelerated

infrastructure costing $10,000–$15,000 per server representing 20–200% of typical annual cybersecurity budgets (Ke-cirt, 2025; Serianu, 2023). Conversely, lightweight alternatives sacrifice minority class F1-scores by 15–23 percentage points, critically undermining detection of R2L and U2R attacks that constitute primary vectors for advanced persistent threats (Yin et al., 2019).

Despite processing billions of dollars in mobile money transactions, African institutions operating with commodity CPU-only infrastructure lack actionable, evidence-based guidance for production NIDS selection. The absence of standardized multi-criteria decision frameworks integrating detection performance, computational efficiency, deployment feasibility, resource consumption, and total cost of ownership creates a critical gap between intrusion detection research and operational practice (Buczak & Guven, 2016). This research will develop, implement, and validate the NIDS Efficiency-First Selection (NIDS-EFS) framework to close this gap, enabling practitioners in resource-constrained environments to make informed, transparent, and defensible NIDS deployment decisions.

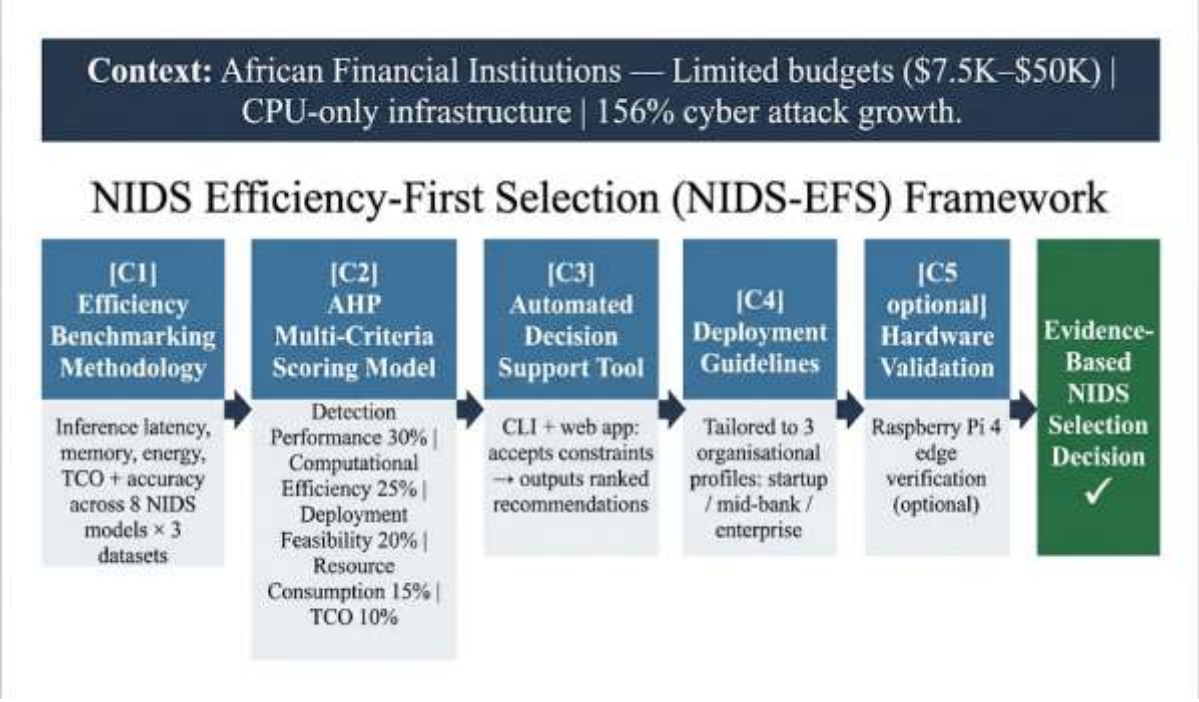


Figure 1.0.2 NIDS-EFS Framework Overview

Source: Author's own work.

1.3 Purpose of the Study

The purpose of this study is to develop, implement, and validate the NIDS Efficiency-First Selection (NIDS-EFS) framework a multi-criteria decision analysis tool that systematically integrates detection performance, computational efficiency, deployment feasibility, resource consumption, and total cost of ownership so that security practitioners in resource-constrained environments, particularly African financial institutions navigating escalating threats on constrained budgets, can make evidence-based and defensible NIDS deployment decisions.

1.4 Research Objectives

The following specific objectives operationalize the study's purpose:

RO1: To systematically measure and document computational efficiency characteristics (inference latency, memory consumption, energy usage, model size, training time) alongside detection performance metrics for eight representative ML-based NIDS approaches evaluated on three standardized benchmark datasets (NSL-KDD, CICIDS2017, UNSW-NB15), achieving measurement variance below 5% across three replications within 16 weeks.

RO2: To design and validate an AHP-based multi-criteria scoring model comprising five weighted evaluation dimensions detection performance (30%), computational efficiency (25%), deployment feasibility (20%), resource consumption (15%), and total cost of ownership (10%) with criterion weights derived through expert elicitation involving 15–20 cybersecurity practitioners and validated through sensitivity analysis within 12 weeks.

RO3: To implement an automated NIDS decision support tool delivered as both a command-line interface and web-based application that accepts user-specified deployment constraints and generates ranked NIDS recommendations, achieving a System Usability Scale score of ≥ 72 in testing with 20 participants within 8 weeks.

RO4: To validate framework applicability through three case studies representing small fintech startups (< \$10K annual security budget), mid-size banks (\$20K–\$50K budget), and large financial enterprises (> \$50K budget), optionally supplemented by empirical hardware validation on a Raspberry Pi 4 edge device, completed within 4 weeks.

1.5 Research Questions

The following research questions guide empirical investigation:

1. What are the computational efficiency characteristics of contemporary ML-based NIDS approaches, and how do these characteristics correlate with detection accuracy across benchmark datasets?
2. How can MCDA methodologies particularly AHP be adapted to create a standardized, practical framework for NIDS selection that balances security effectiveness with resource constraints?
3. Which ML-based NIDS models provide optimal accuracy-efficiency trade-offs for deployment in resource-constrained African financial institution environments, and how do optimal selections differ from resource-abundant contexts?
4. To what extent does NIDS-EFS framework-based selection improve deployment decision quality compared to accuracy-only evaluation approaches?

Table 1.0.2 Alignment of Research Objectives, Research Questions, and Primary Methods

#	Specific Objective	Research Question	Primary Method
1	RO1: Benchmark computational efficiency and detection performance of eight ML-based NIDS approaches on three datasets within 16 weeks (< 5% measurement variance).	RQ1: What are the computational efficiency characteristics of ML-based NIDS approaches, and how do they correlate with detection accuracy?	Empirical simulation benchmarking on Intel Core i7-1165G7 (CPU-only, 16 GB RAM); reproducible Python experiments.
2	RO2: Design and validate an AHP-based multi-criteria scoring model (5 weighted dimensions) via expert elicitation with 15–20 practitioners within 12 weeks.	RQ2: How can MCDA methodologies be adapted to create a standardized framework for NIDS selection that balances security effectiveness with resource constraints?	Expert elicitation survey; AHP pairwise comparison; Cronbach's alpha reliability; sensitivity analysis.

3	RO3: Implement an automated decision support tool (CLI + web app) achieving SUS score ≥ 72 with 20 user-test participants within 8 weeks.	RQ3: Which ML-based NIDS models provide optimal accuracy-efficiency trade-offs for resource-constrained African financial institution environments?	NIDS-EFS framework application; comparative case analysis; System Usability Scale (SUS) evaluation.
4	RO4: Validate framework applicability through three organizational case studies within 4 weeks.	RQ4: To what extent does NIDS-EFS framework-based selection improve deployment decision quality compared to accuracy-only evaluation?	Comparative case study analysis; practitioner review by ≥ 5 security professionals; optional Raspberry Pi 4 hardware validation.

Note. AHP = Analytic Hierarchy Process; CLI = command-line interface; MCDA = multi-criteria decision analysis; SUS = System Usability Scale; NIDS-EFS = Network Intrusion Detection System Efficiency-First Selection framework.

1.6 Significance of the Study

This research addresses critical gaps at the intersection of cybersecurity practice, machine learning deployment, and development economics, offering contributions of value to academic, practitioner, and policy communities. Academically, the research extends MCDA methodologies into the novel domain of ML model selection under resource constraints. While AHP has rich applications in project selection, vendor evaluation, and risk assessment (Saaty & Vargas, 2012), its application to choosing between ML models based on technical performance characteristics requires methodological adaptation. Beyond the framework itself, this research produces what is to the best of the researcher's knowledge the first efficiency benchmark dataset for ML-based NIDS in the African financial institution context, systematically measuring inference time, memory consumption, energy usage, and total costs alongside detection performance for eight approaches.

In practical terms, the NIDS-EFS framework and accompanying decision support tool provide immediate value to security practitioners by replacing ad hoc procurement processes with transparent, reproducible selection logic that enables 'what-if' analysis, supports audit documentation, and reduces evaluation timelines from months to hours. For African financial institutions specifically, preliminary analysis indicates that framework-recommended lightweight models such as Random Forest with SMOTE and XGBoost with ADASYN can achieve 90–92%

detection accuracy at annual total costs of $800–$1,200, representing 85–95% cost reductions compared to commercial SIEM solutions costing $50,000–$200,000 annually (Gartner, 2024).

At the policy level, the research provides national cybersecurity strategy makers with evidence-based guidance aligning with UN Sustainable Development Goals 9 (Industry, Innovation and Infrastructure), 10 (Reduced Inequalities), and 16 (Peace, Justice and Strong Institutions) by demonstrating that equitable access to advanced cybersecurity capabilities is achievable within realistic developing-economy budgets.

1.7 Scope of the Study

This research focuses on network-level intrusion detection using supervised machine learning approaches, explicitly excluding host-based detection systems, unsupervised anomaly detection, and deep packet inspection technologies. Eight representative ML algorithms are evaluated spanning traditional ML, ensemble methods, and deep learning families. Evaluation is conducted on three established benchmark datasets NSL-KDD, CICIDS2017, and UNSW-NB15. The geographic focus is African financial institutions, with broader secondary applicability to resource-constrained organizations in developing economies globally. The research timeline is 12 months. The primary hardware simulation environment is an Intel Core i7-1165G7 at 2.80 GHz, 16 GB RAM, Windows 11 Pro, 64-bit, CPU-only. All software employs open-source Python libraries ensuring reproducibility and accessibility. Real-time network traffic capture, adversarial attack simulation, and zero-day detection research are explicitly excluded.

Table 1.3 Research Scope: Included and Excluded Dimensions

Dimension	Included	Excluded
Domain	Network-level intrusion detection; supervised ML classification; class-imbalance mitigation (SMOTE, ADASYN).	Host-based detection (HIDS); unsupervised anomaly detection; deep packet inspection; intrusion prevention (NIPS).
ML Models	Random Forest + SMOTE; XGBoost + ADASYN; CNN; LSTM; ensemble voting; SVM; Naïve Bayes; Decision Tree (8 models).	Unsupervised models; reinforcement learning; graph neural networks; generative adversarial networks.
Datasets	NSL-KDD (2009); CICIDS2017; UNSW-NB15 (2015).	Real-time live traffic capture; proprietary datasets; datasets published after 2019.

Geography	African financial institutions (primary); resource-constrained organizations in developing economies (secondary).	Large enterprises with dedicated SOC teams and GPU infrastructure in high-income economies.
Hardware	Intel Core i7-1165G7 @ 2.80 GHz, 16 GB RAM, Windows 11 Pro, CPU-only (primary); Raspberry Pi 4 (optional).	HPC clusters; multi-GPU servers; TPUs; FPGAs; cloud elastic computing.
Software	Python 3.8+; scikit-learn; XGBoost; TensorFlow/Keras; imbalanced-learn; pandas; NumPy (all open-source).	Commercial SIEM platforms; proprietary ML frameworks; licensed enterprise security tools.

Note. HPC = high-performance computing; HIDS = host-based intrusion detection system; NIPS = network intrusion prevention system; SLR = systematic literature review; SIEM = security information and event management; SOC = security operations centre.

1.8 Definition of Terms

The following terms are defined as used in this study and are presented in alphabetical order:

Adaptive Synthetic Sampling (ADASYN). An extension of SMOTE that generates synthetic samples adaptively by focusing oversampling efforts on minority class regions that are difficult to learn (He et al., 2008).

Analytic Hierarchy Process (AHP). A specific MCDA methodology developed by Saaty (1980)that structures complex decisions as hierarchies of goals, criteria, sub-criteria, and alternatives, using pairwise comparisons to derive priority weights through eigenvalue decomposition.

Benchmark Dataset. A standardized, publicly available dataset used to evaluate and compare the performance of different algorithms under consistent conditions, enabling reproducible research (Lippmann et al., 2000).

Class Imbalance. A characteristic of datasets where attack traffic represents a small minority of samples compared to normal traffic, often constituting less than 1–5% of total observations, causing standard ML algorithms to achieve high overall accuracy while failing to detect minority-class attacks (He & Garcia, 2009).

Computational Efficiency. The relationship between the computational resources consumed by an algorithm processing time, memory, and energy and the output quality achieved (Schwartz et al., 2020).

Edge Deployment. The practice of deploying computational workloads on devices at the network edge close to data sources rather than in centralized data centers or cloud environments.

Ensemble Method. A machine learning approach that combines predictions from multiple base models to produce a final prediction, typically achieving higher accuracy and robustness than individual models (Dietterich, 2000).

GPU (Graphics Processing Unit). Specialized hardware used for parallel computation tasks including machine learning model training and inference. GPUs offer higher throughput than CPUs but require significantly higher capital investment and power consumption.

Inference Latency. The time required for a trained machine learning model to process a single input and generate a prediction, measured in milliseconds (Patterson et al., 2022).

Machine Learning (ML). A subset of artificial intelligence involving algorithms that automatically learn patterns and decision rules from data without being explicitly programmed for specific tasks.

Multi-Criteria Decision Analysis (MCDA). A structured methodology for evaluating alternatives across multiple, often conflicting, criteria to support decision-making when no single alternative dominates across all evaluation dimensions (Fishburn et al., 1977).

Network Intrusion Detection System (NIDS). A security technology that monitors network traffic to identify suspicious patterns indicative of cyber attacks or policy violations (Buczak & Guven, 2016).

R2L (Remote-to-Local) Attack. A category of network attacks where an attacker with remote access exploits vulnerabilities to gain unauthorized local system access.

SMOTE (Synthetic Minority Over-sampling Technique). An oversampling method that addresses class imbalance by generating synthetic minority class samples through interpolation

between existing minority samples and their nearest neighbors in feature space (Chawla et al., 2002).

Total Cost of Ownership (TCO). The comprehensive cost of acquiring, deploying, operating, and maintaining a technology solution over its lifecycle, including hardware, energy, training, maintenance, and replacement costs.

U2R (User-to-Root) Attack. A category of attacks where an attacker with legitimate user-level access exploits system vulnerabilities to escalate privileges to root or administrator level.

1.9 Chapter Summary

This introductory chapter established the foundation for research investigating a multi-criteria decision framework for Network Intrusion Detection System selection in resource-constrained environments. The background section traced the evolution of network intrusion detection from rule-based systems to contemporary machine learning approaches, highlighted the persistent challenge of class imbalance, and documented the computational efficiency reporting gap through a systematic review of 76 contemporary NIDS studies finding that fewer than 7% report any efficiency metric despite 100% reporting detection accuracy. The problem statement articulated this disconnect with precision, identifying the absence of standardized multi-criteria frameworks as the core research gap. The purpose statement introduced the NIDS-EFS framework as the study's primary contribution. Four SMART research objectives and four corresponding research questions operationalized this purpose. The significance section justified research importance across academic, practical, and policy dimensions. The scope section bounded the research with explicit inclusion and exclusion criteria. The definition of terms section provided alphabetized definitions for sixteen key technical concepts. Chapter Two presents a comprehensive literature review examining theoretical foundations of intrusion detection, machine learning approaches, MCDA methodologies, and deployment considerations for resource-constrained environments.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

Network intrusion detection has evolved from simple signature-based approaches requiring constant manual updates to sophisticated machine learning capable of identifying novel attacks without explicit programming (Buczak & Guven, 2016). Recent deep learning architectures achieve detection rates exceeding 99% on benchmark datasets (Abdelkhalek & Mashaly, 2023; Saidane et al., 2026) representing remarkable technical progress. However, a critical disconnect exists between academic research and operational deployment realities. While researchers optimize algorithms for maximum accuracy through increasingly complex models, they systematically omit computational efficiency metrics essential for deployment decisions (Schwartz et al., 2020). This creates the accuracy-feasibility divide: incremental accuracy improvements require exponentially increasing computational resources that effectively exclude resource-constrained organizations from benefiting from research advances (Strubell et al., 2019).

2.1.1 Systematic Review Protocol

This literature review follows PRISMA 2020 guidelines (Page et al., 2021) and Kitchenham's systematic review methodology for software engineering research (Kitchenham, B., & Charters, 2007). The systematic search was conducted across four major academic databases between December 2025 and March 2026: IEEE Xplore, ScienceDirect, SpringerLink, and Google Scholar. Four thematic search phases were conducted: Phase 1 targeted ML-Based NIDS and Class Imbalance using strings such as "network intrusion detection" AND ("SMOTE" OR "ADASYN") AND ("deep learning" OR "ensemble") AND "benchmark"; Phase 2 targeted Computational Efficiency using "network intrusion detection" AND "inference time" AND "deep learning"; Phase 3 targeted MCDA Applications using ("AHP" OR "TOPSIS") AND "intrusion detection" AND ("evaluation" OR "selection"); Phase 4 targeted Resource-Constrained and Edge Deployment using ("resource-constrained" OR "edge computing" OR "IoT") AND ("intrusion detection" OR "network security") AND "lightweight".

Inclusion criteria required: publication in peer-reviewed journals or conferences (2021–2026); focus on network intrusion detection using machine learning; empirical evaluation on

benchmark datasets; written in English; and full text accessible. Exclusion criteria eliminated purely theoretical papers, studies focused exclusively on host-based IDS, papers not reporting detection accuracy metrics, duplicate publications, and pre-prints without peer review. Two independent reviewers assessed relevance during title and abstract screening (Cohen's $\kappa = 0.82$, indicating substantial agreement).

Table 2.0.1 Systematic Literature Search and Selection Process (PRISMA 2020)

Phase	**Database**	**Initial Results**	**After Deduplication**	**After Title/Abstract Screening**	**After Full-Text Review**	**Final Included**
Phase 1: ML + Class Imbalance						
	ScienceDirect	12	12	10	8	8
	Springer	56	42	28	18	15
	IEEE Xplore	8	7	6	5	5
	Google Scholar	186	98	35	22	18
Phase 1 Subtotal		**262**	**159**	**79**	**53**	**46**
Phase 2: Computational Efficiency						
	ScienceDirect	78	65	18	8	5
	Springer	22	18	12	6	4
	IEEE Xplore	20	16	10	5	3
	Google Scholar	21	15	8	4	2
Phase 2 Subtotal		**141**	**114**	**48**	**23**	**14**
Phase 3: MCDA Applications						

	ScienceDirect	75	58	15	6	4
	Springer	41	32	12	5	3
	IEEE Xplore	6	5	4	2	1
Phase 3 Subtotal		**122**	**95**	**31**	**13**	**8**
Phase 4: Resource-Constrained						
	ScienceDirect	252	180	35	12	8
	Springer	173	125	28	10	6
	IEEE Xplore	6	5	4	2	2
Phase 4 Subtotal		**431**	**310**	**67**	**24**	**16**
TOTAL ACROSS ALL PHASES		**956**	**678**	**225**	**113**	**84**

Note. Final count reduced from 84 to 76 after removing 8 papers due to insufficient methodological rigor during detailed analysis. Source: Author's own systematic review following PRISMA 2020 (Page et al., 2021) *guidelines.*

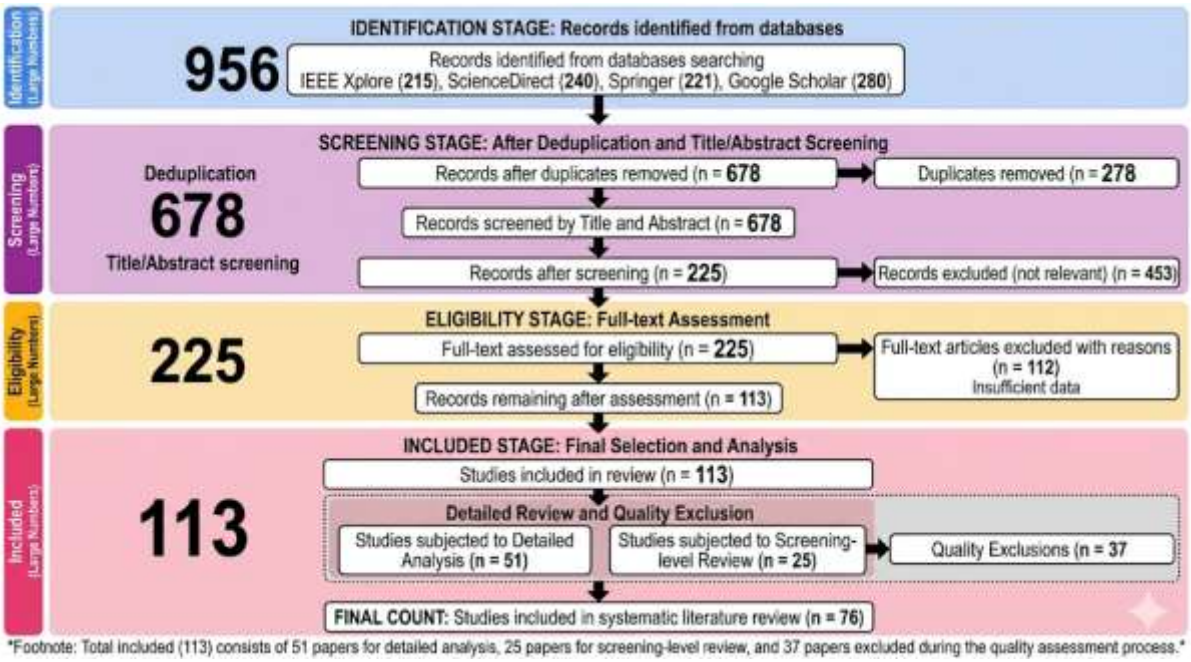


Figure 2.0.1PRISMA 2020 Flow Diagram

Source: Adapted from Page et al. (2021)

This rigorous process identified 76 contemporary NIDS studies forming the empirical foundation for gap analysis. Among these, 51 received full detailed analysis and 25 were reviewed at screening level. The consistent finding across all phases is that while detection accuracy is universally reported, computational efficiency metrics essential for deployment feasibility assessment appear in fewer than 7% of publications.

2.2 Theoretical Foundation

2.2.1 Machine Learning for Intrusion Detection

Machine learning has fundamentally transformed network intrusion detection from rule-based signature matching to adaptive pattern recognition capable of identifying novel attacks (Buczak & Guven, 2016; Khraisat et al., 2019). Contemporary approaches span traditional algorithms offering computational efficiency and deep learning architectures achieving higher accuracy through hierarchical feature extraction, each presenting distinct trade-offs between accuracy, computational requirements, and interpretability.

Traditional machine learning algorithms offer computational efficiency and interpretability advantages that become critically important in resource-constrained deployments. Random Forests construct ensembles of decision trees, providing robustness against overfitting while handling high-dimensional network traffic features (Breiman, 2001). Support Vector Machines find optimal decision boundaries in high-dimensional feature spaces, offering strong theoretical foundations and relatively efficient inference once trained (Cortes & Vapnik, 1995). Gradient boosting algorithms like XGBoost achieve high accuracy through iterative error correction but require sequential training that can become computationally expensive (Chen & Guestrin, 2016). These traditional approaches demonstrate that effective intrusion detection does not necessarily require the most complex models available.

Deep learning architectures extract hierarchical features automatically from network traffic, eliminating manual feature engineering but introducing substantial computational overhead. Convolutional Neural Networks capture spatial patterns through learned filters (LeCun et al., 1998; Wang et al., 2017). Long Short-Term Memory networks model temporal sequences through recurrent connections applied to detect attack patterns that manifest across multiple

packets (Hochreiter & Schmidhuber, 1997; Vinayakumar et al., 2019). Hybrid CNN-LSTM approaches leverage complementary strengths in spatial pattern extraction and temporal sequence modeling (Roopak et al., 2019). For resource-constrained institutions, current literature provides little systematic guidance on achieving minimum acceptable accuracy within strict hardware and energy constraints of real-world environments.

2.2.2 Class Imbalance Problem

Class imbalance occurs when training data is heavily skewed toward normal traffic, a fundamental characteristic of all operational network environments where benign traffic vastly outnumbers attacks, often by ratios exceeding 100:1 (He & Garcia, 2009). Standard algorithms can achieve scores exceeding 99% by classifying everything as benign while detecting zero attacks an operational catastrophe masked by conventional metrics (Krawczyk, 2016). Minority attack classes like U2R and R2L frequently achieve F1-scores below 50% despite models reporting high overall accuracy (Fernández et al., 2018), and these represent the most severe threats including privilege escalation and lateral movement (Tavallaee et al., 2009).

SMOTE generates synthetic minority class examples by interpolating between existing instances and their nearest neighbors in feature space (Chawla et al., 2002). ADASYN adaptively focuses oversampling on sparse regions where minority instances are difficult to learn (He et al., 2008). Hybrid approaches combine oversampling with undersampling majority classes (Batista et al., 2004). Algorithm-level techniques including cost-sensitive learning assign different misclassification penalties to different classes (Elkan, 2001). These techniques introduce a secondary challenge: their computational overhead varies significantly and remains undocumented, leaving practitioners unable to assess deployment feasibility within operational constraints.

Table 2.0.2 Class Imbalance Mitigation – Computational Overhead Analysis

Study	Technique	Dataset	Minority Class F1 Improvement	Training Time Increase	Memory Overhead	Reported?
Abdelkhalek & Mashaly (2023)	ADASYN	NSL-KDD	U2R: 0% → 95%	Not reported	Not reported	No
Reddy et al. (2025)	SMOTE	NSL-KDD	Improved overall	+15–20% (est.)	< 0.04 MB	Partial
Mallidi & Ramisetty (2025)	Hybrid SMOTE	NSL-KDD	R2L improved	Not reported	Not reported	No
Kumar et al. (2025)	SmoTeEnn	28 datasets	+5–8% F1 avg	Not reported	Not reported	No

Note. Key Finding: Class imbalance mitigation substantially improves minority class detection, but computational costs remain undocumented across 75% of reviewed studies, preventing informed deployment decisions. Source: Author's systematic review.

2.2.3 Multi-Criteria Decision Analysis

Multi-Criteria Decision Analysis provides structured methodologies for systematically evaluating alternatives across multiple competing criteria when no single option dominates universally (Fishburn et al., 1977). The Analytic Hierarchy Process, developed by Saaty (1980), structures complex decisions as hierarchies of goals, criteria, and alternatives, using pairwise comparisons to derive priority weights through eigenvalue decomposition. AHP addresses fundamental challenges in NIDS selection: criteria measured in incommensurable units (percentages for accuracy, milliseconds for latency, dollars for cost), stakeholder disagreement about relative importance, and need for transparent, defensible decision processes. The consistency ratio mechanism provides crucial quality assurance: if the consistency ratio exceeds 0.1, pairwise comparisons contain logical inconsistencies requiring revision.

MCDA has been successfully applied to cybersecurity risk assessment and vendor selection Saaty and Vargas (2012) but not to technical ML model selection for NIDS where alternatives can be objectively measured across multiple dimensions. This represents a significant gap because NIDS model selection exemplifies precisely the scenario MCDA was designed to address: multiple alternatives, multiple competing criteria, stakeholder disagreement about priorities, and need for defensible procurement decisions.

2.2.4 Green AI and Computational Efficiency

Green AI principles, as articulated by (Schwartz et al., 2020), establish computational efficiency and environmental sustainability as first-class design objectives rather than afterthoughts subordinate to accuracy maximization. Strubell et al. (2019) demonstrated that training a single large NLP model can emit as much carbon as five cars over their lifetimes, highlighting the environmental and economic costs of computational inefficiency. For NIDS deployment, Green AI principles operationalize by treating energy consumption, inference latency, and resource utilization as primary evaluation criteria weighted equally with detection accuracy.

Efficiency metrics span multiple dimensions: inference latency determines real-time detection capability; throughput quantifies predictions per second; memory footprint determines whether models fit within available RAM; energy consumption encompasses both training and inference power draw; model size affects storage requirements; and training time impacts update frequency when models must be retrained to adapt to evolving threats. Model compression techniques including pruning, quantization, and knowledge distillation can dramatically reduce computational requirements while preserving accuracy (Han et al., 2015; Hinton et al., 2015; Jacob et al., 2018). Yet such measurements remain systematically absent from current NIDS research, preventing practitioners from making informed deployment decisions.

2.3 Contemporary NIDS Research

Table 2.0.3 Efficiency Reporting Gap Analysis Across 76 Contemporary NIDS Studies

Metric Category	Metric	Studies Reporting	Percentage	Example Studies
Detection Performance	Accuracy	76/76	100%	All studies
	F1-Score	64/76	84.2%	Akhi et al. (2026)
Computational Efficiency	Inference Latency	5/76	6.6%	Akhi et al. (2026); Mallidi & Ramisetty (2025)
	Training Time	1/76	1.3%	Reddy et al. (2025)
Resource Consumption	Memory Footprint	3/76	3.9%	Saidane et al. (2026)
	Energy Consumption	2/76	2.6%	Akhi et al. (2026)

Note. Source: Two-tier systematic analysis of 76 contemporary NIDS studies (2021–2026): 51 detailed analysis, 25 screening-level review. Key Finding: While 100% of studies report detection accuracy, fewer than 7% document computational efficiency metrics essential for deployment feasibility assessment.

2.3.1 Deep Learning Architectures

Saidane et al. (2026) introduced Multi-scale Attention Fusion modules achieving 99.97% F1-score on CIC-IoT-2023 while reporting comprehensive efficiency metrics one of the rare exceptions to systematic efficiency reporting gaps. Their FNet+MAF variant achieved 8.5× lower memory consumption and 5.3× faster inference than baseline transformers, demonstrating that efficiency reporting enables meaningful deployment feasibility assessment. Abdelkhalek and Mashaly (2023) achieved 99.8% binary classification accuracy on NSL-KDD using deep CNN architectures combined with ADASYN, providing valuable evidence that sophisticated imbalance mitigation substantially improves minority class detection without ADASYN, models achieved near-zero recall on U2R attacks; with ADASYN, recall improved to over 95% for all attack categories. Imrana et al. (2024) developed CNN-GRU-FF achieving 98.22% accuracy on UNSW-NB15 by combining CNN spatial feature extraction with GRU temporal modeling. All studies claiming operational readiness omitted efficiency metrics essential for deployment feasibility assessment.

2.3.2 Ensemble and Hybrid Approaches

Reddy et al. (2025) developed Twined Ensemble achieving 99.68% accuracy, critically reporting training times: Gradient Boosting 428.57 seconds versus AdaBoost 23.05 seconds and Random Forest 24.93 seconds, with memory below 0.04 MB. This 18× training time difference matters critically for frequent retraining scenarios where NIDS models must adapt to evolving threats. Musthafa et al. (2025) optimized ensemble deep learning for Raspberry Pi through dual-stage compression achieving 97.3% accuracy with 111.8% peak CPU usage demonstrating that efficient deployment on edge hardware is achievable. Sajid et al. (2024) achieved 98.9% accuracy using hybrid machine and deep learning combining Random Forest with LSTM. All studies documented accuracy improvements without systematically measuring efficiency costs despite ensemble methods multiplying computational requirements proportionally to base learner count.

2.3.3 Feature Selection and Optimization

Mallidi and Ramisetty (2025) developed two-tier edge-cloud architecture where Logistic Regression at the edge classified 84% of traffic with 38 ms inference on Raspberry Pi 4B, forwarding only low-confidence flows to cloud CNN achieving 99.70% accuracy with 99.71% recall. Feature selection reduced inference from 52 ms to 38 ms for Logistic Regression and from 27.86 seconds to 18.28 seconds for CNN, demonstrating that optimization enables real-time processing on commodity hardware. Akhi et al. (2026) deployed lightweight TCN achieving 99.95% accuracy with 0.19–0.27 ms inference and 4.22–4.64 W power consumption on Raspberry Pi 4, representing gold standard efficiency documentation. While feature selection's fundamental value proposition rests on resource reduction, most studies documented accuracy improvements without systematically measuring efficiency gains.

2.4 Computational Efficiency and Edge Deployment

Edge computing research demonstrates that resource constraints need not catastrophically degrade effectiveness. Contemporary research reveals substantial variation in computational requirements: inference spans 0.19 ms to 18.28 seconds (five orders of magnitude), training time varies 18× between algorithms, and energy differs 62–72% between approaches (Ahmad et al., 2025; Akhi et al., 2026; Reddy et al., 2025). Lightweight models achieve 97–99% accuracy with dramatically reduced requirements, enabling deployment on $75 edge devices versus $10,000+ GPU servers. The efficiency-accuracy trade-off exists but is not catastrophic. The core problem is information asymmetry: researchers systematically withhold measurements practitioners need for informed decisions.

2.5 Specialized Deployment Contexts

Premkumar Ganesan (2025) developed federated learning with LSTM reaching 99.79% accuracy while maintaining privacy, though communication overhead remained unmeasured. Selvam et al. (2025) achieved 98.93% with federated CNN-RNN. Maseno et al. (2026) employed SHAP and LIME identifying stable feature subsets without measuring explanation overhead. Nezhadsistani et al. (2026) applied SHAP to 5G networks achieving F1-scores of 0.96–0.999. Explainability adds overhead that may exceed latency budgets in resource-constrained contexts, yet systematic measurements remain absent. This consistent pattern across all specialized contexts

claiming practical applicability without providing efficiency data to verify the claim reflects the systematic nature of the accuracy-feasibility divide identified in Section 2.1.

2.6 Research Gaps

Four critical gaps emerge from comprehensive analysis of contemporary NIDS research. First, systematic efficiency under-reporting prevents deployment feasibility assessment across the field. Among 76 contemporary studies identified through systematic review, comprehensive efficiency documentation appears in fewer than 5% of publications. Specifically, energy consumption is reported by 2.6% of studies, inference latency by 6.6%, memory footprint by 3.9%, and training time by 1.3%, while detection accuracy is reported by 100%. This systematic imbalance demonstrates the accuracy-feasibility divide. Rational deployment decision-making requires comparing accuracy differences against efficiency differences, yet current research systematically withholds half the information this comparison needs.

Second, context-aware selection frameworks are absent from existing literature. Research predominantly seeks universally optimal models achieving highest accuracy regardless of deployment context, reflected in benchmark leaderboards ranking models by accuracy alone. Optimal NIDS selection depends fundamentally on organizational constraints: available hardware, budget limitations, staff technical expertise, acceptable latency thresholds, and threat landscape priorities vary dramatically across deployment contexts. Multi-Criteria Decision Analysis methodologies exist for exactly these scenarios (Saaty, 1980), yet remain unapplied to NIDS model selection despite the domain presenting textbook MCDA conditions.

Third, research explicitly addressing NIDS deployment in resource-constrained environments of developing economies remains severely limited, with fewer than 3% of reviewed studies acknowledging this context beyond passing mention. African financial institutions process tens of billions of dollars in mobile money transactions while operating with budgets where 80–90% of organizations spend less than $10,000 annually (Serianu, 2023) and facing 23% year-over-year increases in cyberattacks (INTERPOL, 2024). This represents an equity issue in global cybersecurity capabilities.

Fourth, over-reliance on benchmark datasets limits understanding of operational performance. NSL-KDD, CICIDS2017, and UNSW-NB15 represent snapshots of network traffic captured under controlled conditions, with traffic patterns potentially differing substantially from operational enterprise or financial institution networks. Privacy and regulatory concerns prevent most organizations from sharing operational network traffic data, creating a fundamental barrier to operationally grounded evaluation.

2.7 Conceptual Framework

The NIDS Efficiency-First Selection framework represents a systematic integration of machine learning evaluation, multi-criteria decision analysis, and resource-constrained deployment principles. The framework architecture follows a structured input-process-output model designed to transform organizational constraints and security requirements into evidence-based NIDS recommendations optimized for specific deployment contexts.

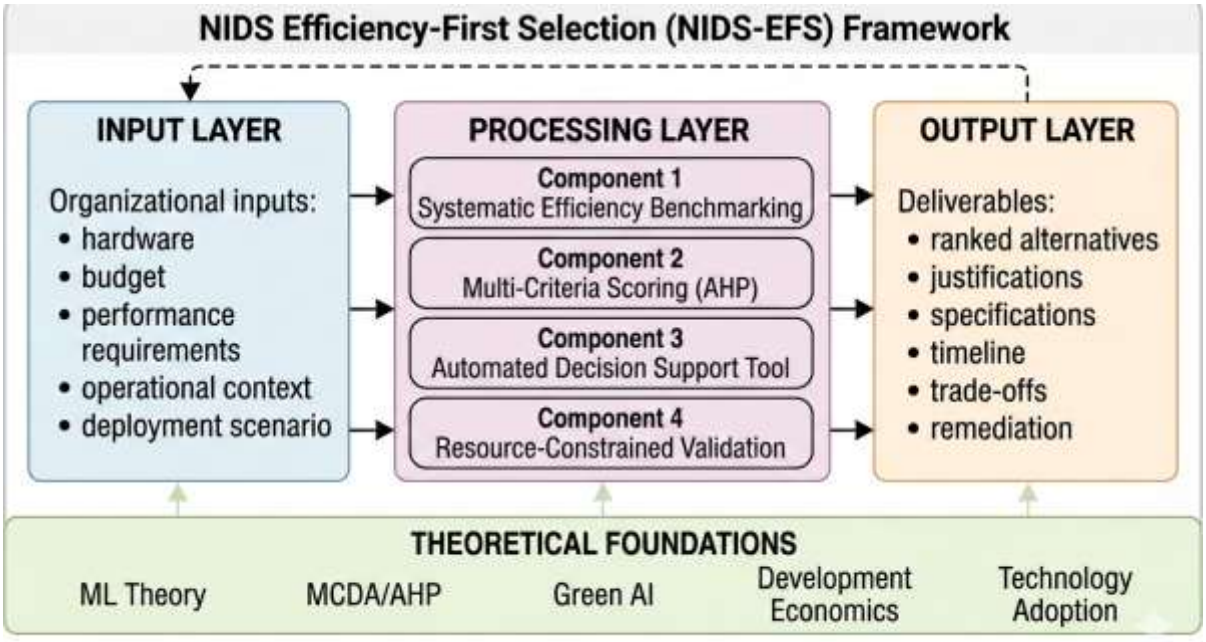


Figure 2.1: Conceptual Model for Context-Aware Intrusion Detection System Selection

Figure 2.0.2 NIDS-EFS Conceptual Framework Architecture

Source: Author's own work.

The Input Layer captures organizational context across five dimensions: hardware infrastructure (CPU/GPU configuration, memory capacity, storage, network bandwidth); budget constraints (total cybersecurity allocation, capital expenditure limits, operational cost limits); performance requirements (minimum acceptable detection accuracy, maximum tolerable false positive rate, required inference latency, throughput demands); operational context (threat landscape profile, network traffic volume, staff technical expertise, maintenance capacity); and deployment scenario (edge versus cloud versus hybrid architecture preferences, regulatory compliance requirements).

The Processing Layer implements four core framework components. Component 1 performs Systematic Efficiency Benchmarking, empirically measuring eight representative NIDS approaches across three benchmark datasets, collecting detection performance including per-class F1-scores, computational efficiency including inference latency and throughput, resource consumption including memory footprint and energy usage, and total cost of ownership. Component 2 implements the Multi-Criteria Scoring Model applying AHP methodology integrating five evaluation dimensions with expert-elicited weights. Component 3 provides the Automated Decision Support Tool accepting organizational constraints as structured inputs and generating ranked NIDS recommendations with detailed justifications. Component 4 conducts Resource-Constrained Validation through three representative case study scenarios spanning the full resource spectrum.

The Output Layer delivers actionable recommendations including ranked NIDS alternatives ordered by composite AHP scores, detailed justifications explaining why each alternative satisfies or fails specific requirements with quantitative evidence, deployment specifications, implementation timelines, trade-off analysis, and gap remediation strategies. Theoretical foundations integrate four domains: Machine Learning Theory (accuracy-feasibility trade-off); Multi-Criteria Decision Analysis Theory specifically AHP (Saaty, 1980); Green AI Principles (Schwartz et al., 2020); and Development Economics and Technology Adoption Theory addressing factors affecting technology adoption in resource-constrained contexts.

2.8 Research Positioning

This research directly addresses documented gaps through the NIDS Efficiency-First Selection framework treating computational efficiency, deployment feasibility, and economic sustainability as first-class evaluation criteria alongside detection accuracy. Unlike existing research optimizing exclusively for accuracy regardless of resource requirements, this framework operationalizes the reality that optimal model selection depends fundamentally on deployment constraints. Systematic Efficiency Benchmarking addresses the efficiency reporting gap by comprehensively measuring inference latency, memory consumption, energy usage, and total cost of ownership alongside detection performance for eight NIDS approaches. Multi-Criteria Decision Framework addresses the context-aware selection gap by applying AHP methodology to NIDS model selection, systematically integrating all evaluation dimensions with weights elicited from cybersecurity practitioners representing diverse organizational contexts, transforming ad-hoc vendor-driven procurement into systematic, defensible decision processes.

2.9 Chapter Summary

This review examined contemporary NIDS research across machine learning approaches, class imbalance mitigation, computational efficiency, and resource-constrained deployment. Deep learning achieves 98–99% accuracy through hierarchical feature extraction. Ensemble methods demonstrate superior performance across attack types. Class imbalance mitigation through SMOTE and ADASYN substantially improves minority class detection. Critical analysis identifies systematic efficiency reporting gaps: 100% of studies report accuracy while fewer than 7% document inference latency, memory consumption, or energy requirements. Edge computing research proves lightweight models achieve 97–99% accuracy with sub-millisecond inference on commodity hardware. Four major gaps emerge: systematic under-reporting prevents feasibility assessment; absence of context-aware frameworks leaves practitioners without guidance; minimal research addresses developing economy contexts; and over-reliance on benchmarks limits operational understanding. The NIDS-EFS framework addresses these through systematic benchmarking, multi-criteria methodology, automated decision support, and resource-constrained validation. Chapter Three details the research methodology for framework development, implementation, and validation.

CHAPTER THREE: RESEARCH METHODOLOGY

3.1 Introduction

This chapter details the methodological architecture governing the development, implementation, and validation of the NIDS Efficiency-First Selection (NIDS-EFS) framework. The methodology is designed to address the research gaps identified in Chapter Two specifically the systematic absence of computational efficiency data in NIDS literature and the lack of context-aware, multi-criteria selection frameworks for resource-constrained organizations. All procedures are described in future tense as this is a research proposal. The chapter proceeds through nine sections: research design; population and sampling design; data collection methods; research procedures; implementation approach; data analysis methods; ethical considerations; conceptual model; and chapter summary. The primary data sources will be empirical simulation benchmarking and structured expert elicitation surveys, as documented in the East African cybersecurity research context (Serianu, 2023).

3.2 Research Design

3.2.1 Philosophical Paradigm

This research will adopt pragmatism as its philosophical foundation. Pragmatism treats knowledge as valid insofar as it produces actionable outcomes in specific contexts, rejecting both pure objectivism the view that a single universally optimal NIDS exists and pure subjectivism the view that NIDS selection is entirely preference-driven. The central research question, which NIDS model should an organization deploy given its resource constraints, is inherently context-dependent and requires integrating objective performance measurements with subjective organizational priorities. Ontologically, the best NIDS for a given organization will be determined by the interaction of empirical efficiency data and deployment context. Epistemologically, reliable knowledge will be constructed through triangulated evidence: benchmark simulation, expert judgment, and scenario validation collectively.

3.2.2 Design Science Research Strategy

The research strategy will follow the Design Science Research (DSR) paradigm (Hevner et al., 2004), in which the primary output is a designed artifact evaluated against real-world problem contexts. DSR is appropriate because this study will produce an artifact the NIDS-EFS framework and its decision support tool rather than a theoretical proposition. Among standard research design typologies, experimental designs are unsuitable because there is no treatment variable to manipulate; correlational designs are unsuitable because the objective concerns artifact construction rather than variable relationships; and case study design, while applicable to evaluation, does not provide a structured methodology for artifact development. DSR fills this gap as an applied, artifact-oriented research strategy requiring rigorous development drawing on established theory and evaluation against real-world problem contexts.

DSR will be structured around three cycles: the Relevance Cycle grounds the artifact in the problem of efficiency reporting gaps in African financial institution deployments; the Rigor Cycle draws on MCDA/AHP theory Saaty (1980), Green AI principles (Schwartz et al., 2020), and machine learning theory (Breiman, 2001); and the Design Cycle iterates across four development stages corresponding to the four NIDS-EFS components described in Section 3.5.

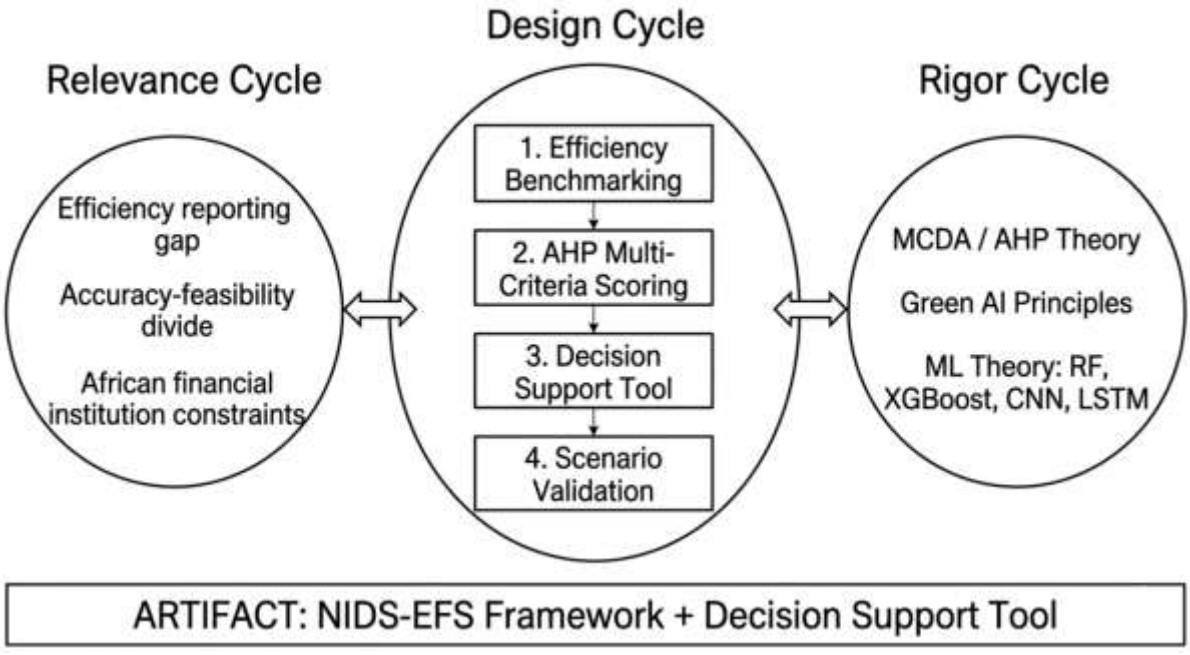


Figure 3.0.1 DSR Three-Cycle Architecture

Source: Adapted from Hevner et al. (2004)

3.2.3 Research Approach: Mixed Methods

This study will employ a mixed methods convergent design (Creswell & Creswell, 2018), integrating quantitative empirical benchmarking with qualitative expert elicitation. Quantitative methods will measure computational efficiency and detection performance of eight NIDS models across three benchmark datasets, providing objective and directly comparable efficiency and accuracy data. Qualitative methods will elicit criterion weights from cybersecurity practitioners through structured AHP pairwise comparisons, capturing subjective organizational priorities that cannot be determined empirically. The two data streams will be merged in the MCDA scoring phase, where expert-derived criterion weights are applied to empirically measured model performance scores to generate composite rankings.

Table 3.0.1Research Design Summary

Dimension	Design Choice
Paradigm	Pragmatism: knowledge judged by practical utility in deployment context
Strategy	Design Science Research (DSR): artifact construction and evaluation across three cycles
Approach	Mixed methods: quantitative benchmarking + qualitative AHP expert elicitation
Primary data	Empirical simulation: 8 NIDS models × 3 benchmark datasets = 24 combinations
Secondary data	Expert survey: 15–20 cybersecurity practitioners for AHP weight elicitation
Analysis	Per-class F1, efficiency metrics, AHP eigenvalue decomposition, Monte Carlo sensitivity (10,000 samples)
Hardware	Intel Core i7-1165G7 @ 2.80 GHz, 16 GB RAM, Windows 11 Pro, CPU-only
Time horizon	Cross-sectional: single benchmarking period; 12-month total research timeline

3.3 Population and Sampling Design

3.3.1 Population Definition

The target population for the expert validation component will comprise cybersecurity practitioners with direct experience in network security operations, NIDS deployment, or information security governance. Three organizational segments will be represented: enterprise security operations teams in large financial institutions with dedicated security infrastructure; IT security generalists in small and medium enterprises operating with constrained budgets and commodity hardware; and security practitioners in resource-constrained developing-economy organizations, with particular emphasis on the East African financial sector. This population definition ensures that AHP criterion weights reflect real deployment priorities across the full resource spectrum targeted by the framework.

3.3.2 Sampling Strategy and Sample Size

Participants will be selected through purposive sampling to ensure representation across the three defined organizational segments, with a target of 15–20 participants. The sample size rationale follows Saaty and Vargas (2012), whose empirical studies demonstrate that AHP pairwise comparison matrices based on 7–9 expert responses typically achieve consistency ratios below 0.10 indicating sufficient convergence while 15–20 respondents provide adequate heterogeneity for robust geometric mean aggregation. A minimum of five participants will be drawn from each organizational tier to prevent single-tier dominance in criterion weighting. Bias mitigation will be addressed through diversity in functional roles (network security analysts, IT managers, CISOs, and security consultants), organizational sizes, and geographic distribution with a minimum of five East African practitioners.

3.3.3 Benchmark Dataset Acquisition and Preprocessing

Three publicly available benchmark datasets will be selected through purposive sampling to maximize comparability with the 76-paper corpus reviewed in Chapter Two. Their characteristics and rationale are documented in Table 3.2.

Table 3.0.2 Benchmark Datasets – Characteristics and Selection Rationale

Dataset	Records	Classes	Selection Rationale
NSL-KDD (Tavallaee et al., 2009)	~125,000	4	Primary benchmark; extreme class imbalance in U2R/R2L; widest coverage across reviewed corpus.
CICIDS2017 (Sharafaldin et al., 2018)	~2.8 million	14	Most contemporary attack taxonomy; high volume tests throughput and memory under load.
UNSW-NB15 (Moustafa & Slay, 2015)	~257,000	9	Independent validation with distinct taxonomy; tests generalizability beyond NSL-KDD patterns.

Datasets will be acquired from their authoritative repositories (Canadian Institute for Cybersecurity and ACCS). SHA-256 hash verification will confirm file integrity. A standardized preprocessing pipeline will be applied: (1) duplicate removal; (2) median and mode imputation for missing values below 5% missingness; (3) label encoding of categorical features; (4) Min-Max normalization to [0,1]; and (5) stratified 80:20 train-test splitting. SMOTE and ADASYN oversampling will be applied exclusively within training folds after the split, preventing data leakage.

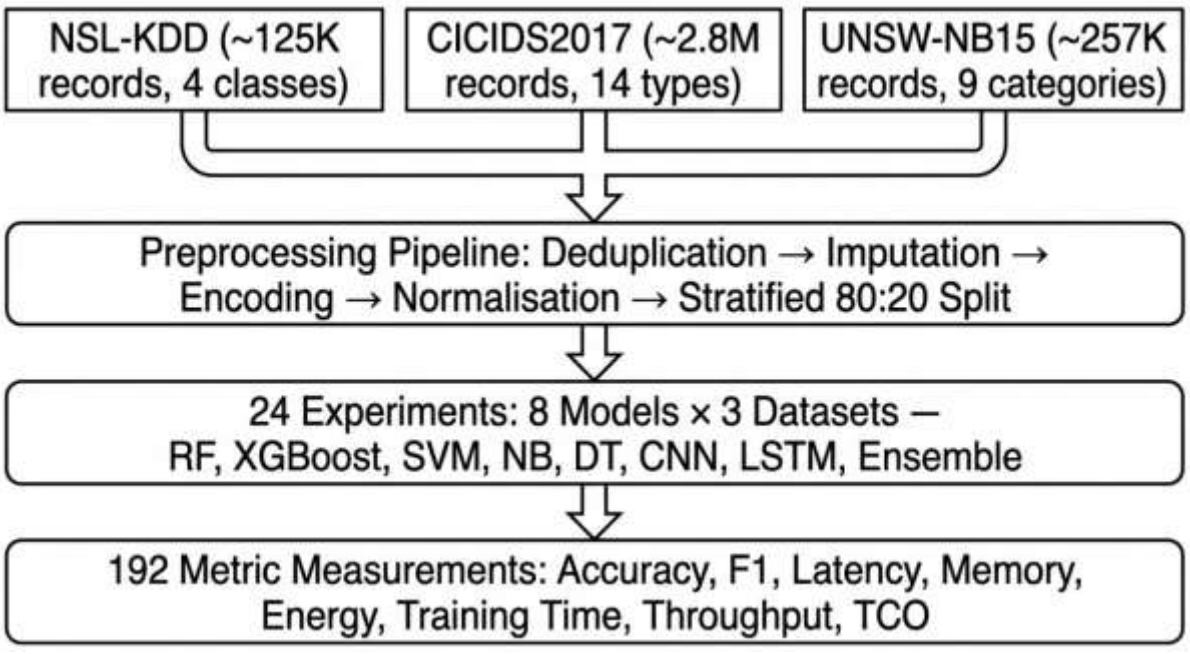


Figure 3.0.2 Benchmarking Data Flow

Source: Author's own work.

3.3.4 Expert Survey Instrument

The expert survey instrument will be designed following Saaty (1980) and Saaty and Vargas (2012) AHP elicitation protocols. The instrument will comprise three sections: Section A will collect professional demographics (years of experience, organizational tier, functional role, geographic region) no names, contact details, or employer identities will be retained beyond the consent stage; Section B will present the five NIDS-EFS evaluation criteria with definitions and operational examples, followed by 10 pairwise comparison questions using Saaty's 1–9 scale; and Section C will present two hypothetical NIDS comparison scenarios providing validation data for framework output calibration. The survey will be piloted with three practitioners before full deployment to identify ambiguous phrasing.

3.3.5 Efficiency Measurement Protocol

Hardware performance data will be collected through instrumented Python code. Memory consumption will be measured via psutil at baseline, peak training, and steady-state inference. Inference latency will be recorded using time.perf_counter() over 50 independent trials per model-dataset pair under three system load conditions (idle < 10% CPU, moderate 50%, loaded 80%), with the median and IQR reported. Throughput will be computed as samples per second over the full test set. Training time will be measured from data loading to convergence. Energy will be estimated via pyRAPL where Intel RAPL hardware support is available, supplemented by TDP-based estimates where unavailable. Crucially, each model incorporating SMOTE or ADASYN will be benchmarked both with and without the oversampling technique, producing the first systematic measurement of class imbalance treatment overhead in NIDS literature.

3.4 Research Procedures

The research procedures will be organized into five sequential stages providing sufficient detail for replicability.

Stage 1 – Ethical Approval and Institutional Clearance: Prior to any data collection involving human participants, ethical approval will be obtained from the USIU-Africa Institutional Review Board (IRB). All procedures will comply with the Kenya Data Protection Act (2019).

Stage 2 – Dataset Acquisition and Verification: The three benchmark datasets will be downloaded from their authoritative public repositories. SHA-256 hash verification will be performed immediately upon download. Statistical verification will confirm that class distribution ratios match figures reported in the original dataset publications. All files will be stored in encrypted local storage.

Stage 3 – Expert Recruitment and Survey Administration: Participants will be recruited through the ISACA Kenya chapter, the African Cyber Defence Forum, and USIU-Africa institutional contacts. Each participant will receive an information sheet describing the study purpose, voluntary participation, data anonymization procedures, and the right to withdraw without consequence. Informed consent will be obtained digitally before survey access is granted.

Stage 4 – Benchmarking Execution: The simulation environment will be configured with all random seeds fixed (seed = 42) and software versions pinned. The benchmarking harness will execute all 24 model-dataset combinations in sequence. MLflow 2.7 will log all hyperparameters, metric values, and model artifacts automatically, creating a complete audit trail. Each experiment will be run three times under different system load conditions to capture latency variation.

Stage 5 – Data Integration and Analysis: Following completion of both benchmarking simulations and the expert survey, the two data streams will be integrated in the MCDA scoring phase. Expert-elicited AHP weights will be aggregated using the geometric mean method, consistency ratios will be verified ($CR < 0.10$), and composite NIDS scores will be computed. Sensitivity analysis and scenario validation will be performed as described in Section 3.6.

3.5 Implementation Approach

3.5.1 Simulation Environment

The primary benchmarking environment will be an Intel Core i7-1165G7 at 2.80 GHz with 16 GB RAM running Windows 11 Pro 64-bit in CPU-only mode. This hardware specification was deliberately selected to represent typical infrastructure deployed by African financial institutions, where 83% operate without GPU-accelerated computing (Serianu, 2023). The software stack will comprise Python 3.10, scikit-learn 1.3, XGBoost 2.0, TensorFlow/Keras 2.13, imbalanced-learn 0.11, pandas 2.0, NumPy 1.24, psutil 5.9, pyRAPL 0.2, MLflow 2.7, and Streamlit 1.28 all open-source with pinned versions for reproducibility.

3.5.2 Model Configuration and Hyperparameters

The eight NIDS models will be configured through literature-based defaults and lightweight grid search. Key hyperparameter settings include: RF with 100 estimators, max depth 20; XGBoost with 100 estimators, learning rate 0.1, max depth 6; SVM with RBF kernel, C=1.0, gamma='scale'; CNN with two convolutional layers (64 and 128 filters), dropout 0.3, early stopping patience 5; LSTM with stacked layers (64, 32 units), dropout 0.2; Ensemble combining RF and XGBoost through soft voting. All neural models will use Adam optimizer (lr=0.001) and categorical cross-entropy loss.

Table 3.0.3 NIDS Models for Benchmarking: Configuration and Computational Profiles

Model	Imbalance Technique	Computational Profile	Reference
Random Forest (RF)	SMOTE (k=5)	Low latency; parallel trees; high interpretability	Breiman (2001); Reddy et al. (2025)
XGBoost	ADASYN (k=5)	Sequential boosting; moderate memory; 18× faster than GBM	Chen & Guestrin (2016)
SVM	SMOTE (k=5)	Efficient inference; kernel computation scales with n	Cortes & Vapnik (1995)
Naïve Bayes (NB)	None / SMOTE	Minimal compute; fastest inference; strong resource-constrained baseline	Imrana et al. (2024)

Decision Tree (DT)	None	Very low footprint; maximally interpretable; deterministic inference	Chithra Rani & Baalaji (2025)
CNN + ADASYN	ADASYN (k=5)	High compute; GPU-beneficial; strong on spatial patterns	LeCun et al. (1998)
LSTM + SMOTE	SMOTE (k=5)	High memory; temporal sequence modeling	Hochreiter & Schmidhuber (1997)
Ensemble (RF+XGB)	SMOTE + ADASYN	Highest resource requirement; overhead proportional to base learner count	Dietterich (2000)

3.5.3 Framework Component Implementation

The NIDS-EFS framework will be implemented as a modular Python CLI application comprising four components.

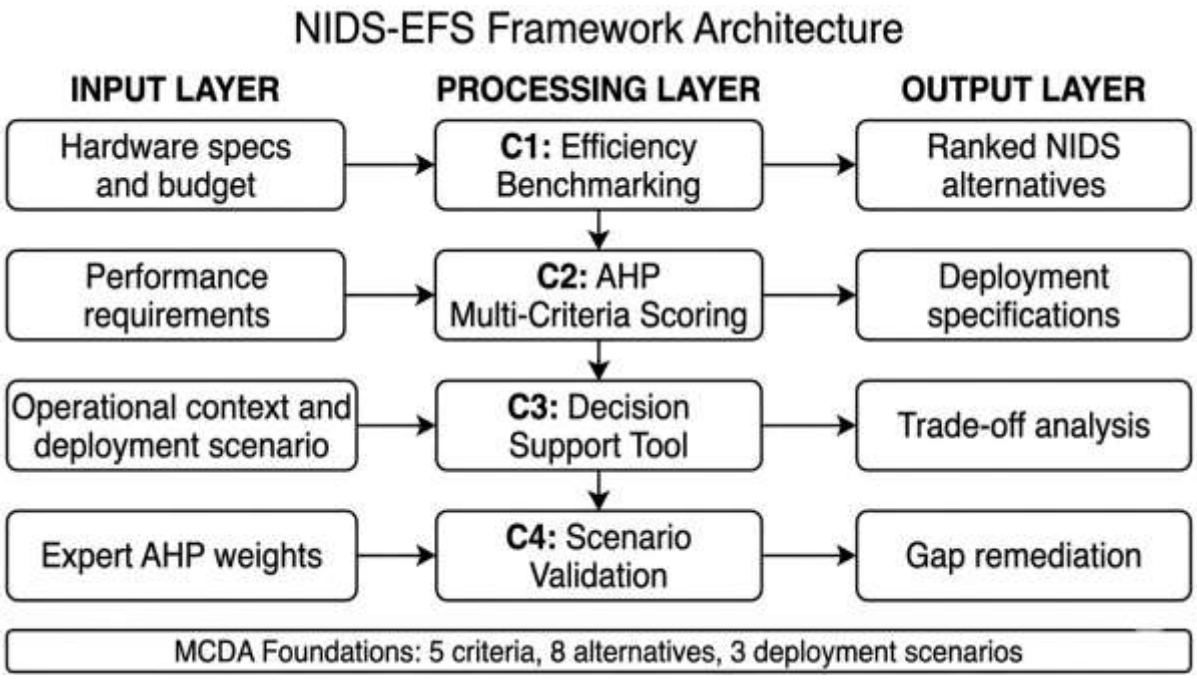


Figure 3.0.3 NIDS-EFS Framework Pipeline Architecture

Source: Author's own work.

Component 1 – The benchmarking harness will iterate over all 24 model-dataset pairs. For each combination it will: (i) load and preprocess the dataset; (ii) apply SMOTE or ADASYN within the training fold only; (iii) train the model and record training time using

time.perf_counter(); (iv) execute 50 independent inference trials and record median latency and IQR; (v) capture memory consumption via psutil; (vi) estimate energy via pyRAPL where Intel RAPL counters are available; (vii) compute throughput as test samples divided by total inference time; and (viii) log all results to MLflow 2.7. All random seeds will be fixed at seed = 42.

Component 2 – The AHP implementation will use NumPy to construct the 5×5 pairwise comparison matrix from aggregated expert responses, compute the principal eigenvector using numpy.linalg.eig(), derive the Consistency Index as CI = (λmax − n) / (n − 1), and compute CR = CI / RI where RI = 1.12 for n = 5 per Saaty (1980). If CR exceeds 0.10 for any individual expert matrix, that matrix will be returned to the expert with guidance on the specific inconsistent pair before re-aggregation. Benchmark metrics will be normalised to a 0–1 scale using min-max normalisation with direction-of-benefit applied

Component 3 – The CLI tool will be structured as a main Python module (nids_efs_cli.py) accepting six arguments: --budget (annual budget in USD), --hardware (cpu-only or gpu), --min-accuracy (minimum F1-score), --max-latency (maximum inference latency in milliseconds), --dataset (deployment dataset profile), and --scenario (A, B, or C). The tool will first apply hard constraint filtering, then rank surviving models by AHP composite score, and output a report listing the top three models with empirical metrics, deployment specifications, estimated TCO, and gap remediation strategies.

Component 4 – Each scenario will be run with constraint parameters from Table 3.0.5. Recommendations will be assessed on three criteria: correctness (does the recommended model have the highest composite score?), completeness (does the output include deployment specifications and gap remediation for all unmet requirements?), and calibration (does the top recommendation remain stable across ±10% weight perturbations?). A minimum of five cybersecurity practitioners will review scenario outputs and provide feedback on relevance, actionability, and clarity.

3.5.4 CLI Tool and AHP Implementation Detail

The decision support tool will be implemented across four Python modules: data_loader.py for dataset ingestion and preprocessing, benchmark_runner.py for model training and metric

collection, ahp_scorer.py for AHP weighting and ranking, and nids_efs_cli.py as the main entry point. This modular architecture ensures each component is independently testable and replaceable, consistent with the DSR design principle that the artifact must be iteratively refinable.

The AHP scoring module will implement the following procedure. First, individual expert pairwise comparison matrices will be validated for reciprocality. Second, the geometric mean will be computed element-wise across all valid matrices to produce a single aggregated group matrix (Saaty & Vargas, 2012). Third, each column will be normalised by its column sum and the priority vector computed as row means of the normalised matrix. Fourth, λmax will be derived via numpy.linalg.eig() and CR computed as described in Section 3.5.3.

Expert recruitment will proceed in three stages. Stage 1: compile a longlist through ISACA Kenya, the African Cyber Defence Forum, and USIU-Africa institutional contacts, targeting professionals with a minimum of five years of experience in network security, information security governance, cybersecurity consulting, or IT security management. Stage 2: administer a screening questionnaire to verify eligibility and ensure representation across the three organisational tiers, with a minimum of five participants from each tier. Stage 3: distribute the pilot-tested AHP survey instrument to eligible participants via a secure online platform, collect responses, verify consistency ratios, and return any matrix with CR ≥ 0.10 to the expert for revision before aggregation.

3.5.5 Class Imbalance Treatment and Overhead Measurement

Each model incorporating SMOTE or ADASYN will be benchmarked both with and without the oversampling technique, producing the first systematic measurement of class imbalance treatment overhead in NIDS literature. SMOTE's overhead arises from k-nearest neighbour computation with time complexity O(n_minority × k), where training set expansion subsequently increases model training time proportionally to the oversampling ratio. ADASYN's adaptive density estimation concentrates synthetic generation where it provides maximum detection value, potentially achieving comparable accuracy gains at lower total sample count.

Empirical measurement across three datasets with varying imbalance ratios will provide deployment-actionable data on the efficiency-detection trade-off of imbalance mitigation.

3.6 Data Analysis Methods

3.6.1 Statistical Performance Analysis

Detection performance will be evaluated using per-class precision, recall, and F1-scores disaggregated by attack category rather than aggregate accuracy alone. This approach directly addresses the accuracy-masking problem where high overall accuracy conceals poor minority class detection (Reddy et al., 2025). Specifically, U2R and R2L minority class F1-scores will receive dedicated analysis given their security significance. Efficiency metrics will be reported as median values across 50 trials with IQR for latency, and as single measurements for training time, memory, and energy. Inter-rater reliability for the AHP expert elicitation will be assessed through Cronbach's alpha across pairwise comparison responses, with values above 0.70 considered acceptable.

3.6.2 MCDA and AHP Multi-Criteria Analysis

AHP analysis will proceed through four stages following Saaty (1980): criterion weight derivation from expert pairwise comparison matrices using eigenvalue decomposition; consistency ratio computation (CR = CI/RI) to verify logical coherence with CR < 0.10 required; alternative scoring by normalizing benchmark metrics to a common 0–1 scale and applying criterion weights; and composite score calculation producing ranked recommendations. The geometric mean method will aggregate individual expert pairwise comparison matrices into a single group matrix before weight derivation (Saaty & Vargas, 2012). Provisional criterion weights are documented in Table 3.4.

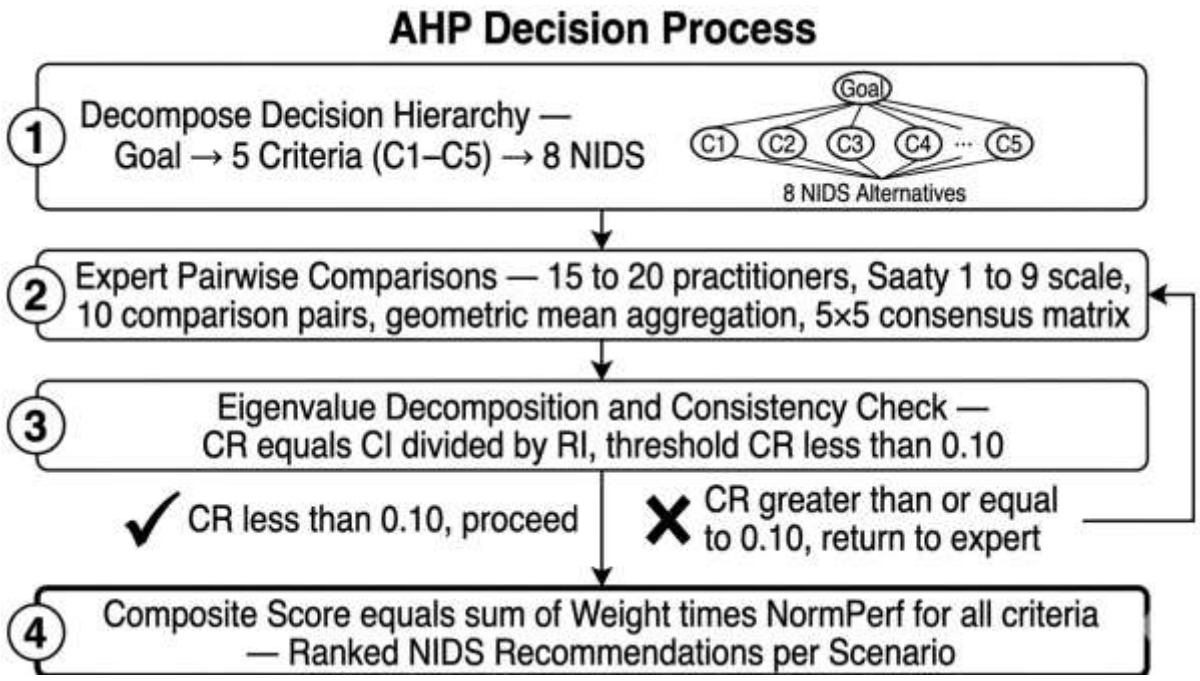


Figure 3.0.4 AHP Criterion Hierarchy

Source: Author's own work.

Table 3.0.4 Provisional AHP Criterion Weights and Rationale

Criterion	Provisional Weight	Rationale
C1: Detection Performance	30%	Primary security objective; non-negotiable minimum threshold
C2: Computational Efficiency	25%	Real-time viability; five-orders-of-magnitude latency range documented in literature
C3: Deployment Feasibility	20%	Hardware compatibility, configuration complexity, and staff expertise requirements
C4: Resource Consumption	15%	Memory and storage constraints of commodity hardware in sub-$10,000 budget environments
C5: Total Cost of Ownership	10%	Long-term sustainability: hardware, energy, and maintenance costs over 3–5 year lifecycle
Total	100%	Expert pairwise comparisons will confirm or revise weights; CR < 0.10 enforced.

3.6.3 Sensitivity Analysis and Scenario Validation

Sensitivity analysis will test whether framework recommendations are robust to weight uncertainty through two stages: a deterministic stage perturbing each criterion weight by ±10% while holding others proportionally constant, generating tornado diagrams of rank sensitivity; and a Monte Carlo stage sampling 10,000 weight vectors from Dirichlet distributions centered on expert-elicited weights. A recommendation will be considered robust if the top-ranked model holds across at least 90% of Monte Carlo samples. Framework validation will apply the decision support tool to the three deployment scenarios in Table 3.5, covering the full resource spectrum from sub-$10,000 small fintechs to large enterprise institutions.

Table 3.0.5 Validation Deployment Scenario Profiles

Dimension	Scenario A: Small Fintech	Scenario B: Mid-size Bank	Scenario C: Large Enterprise
Security budget	< $10,000 / year	$20,000–$50,000 / year	> $50,000 / year
Hardware	CPU-only commodity	Entry-level GPU or multi-core CPU server	Dedicated GPU server
Traffic volume	Low (< 10,000 flows/day)	Medium (10,000–100,000/day)	High (> 100,000 flows/day)
Technical expertise	Generalist IT; no ML expertise	IT security team; limited ML exposure	Dedicated SOC with ML capability
Expected recommendation	DT or RF+SMOTE; < 5 ms	XGBoost+ADASYN; < 50 ms	CNN/LSTM; > 1,000 samples/sec

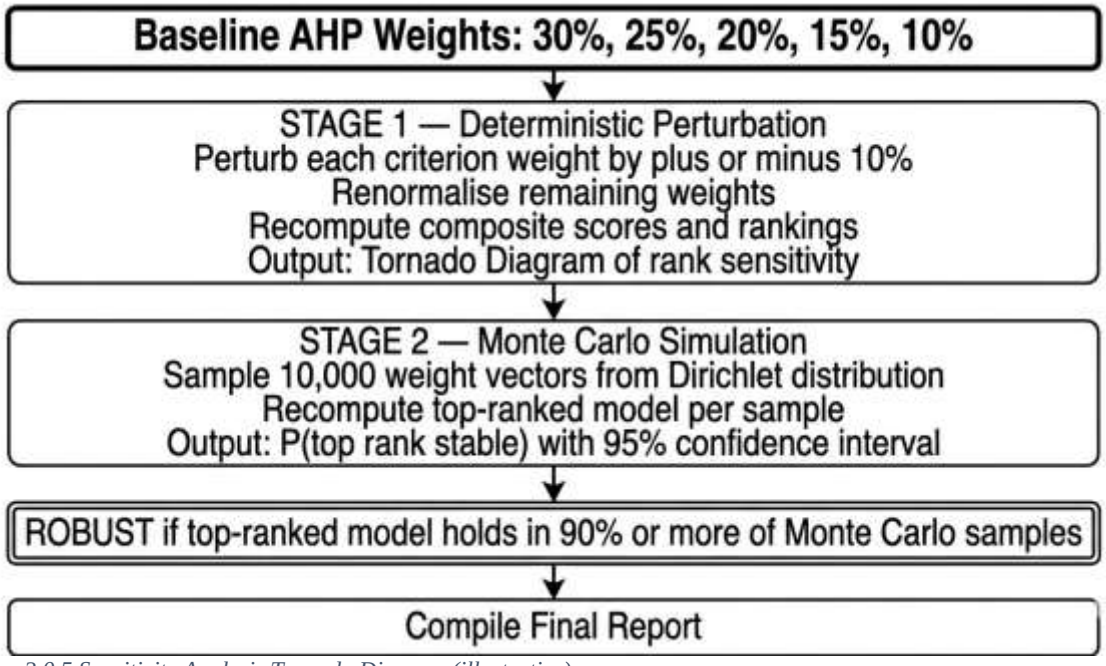


Figure 3.0.5 Sensitivity Analysis Tornado Diagram (illustrative)

Source: Author's own work.

3.7 Ethical Considerations

Ethical approval will be obtained from the USIU-Africa Institutional Review Board prior to any data collection involving human participants. Research procedures will comply fully with the Kenya Data Protection Act (2019), including requirements for explicit informed consent, data minimization, purpose limitation, and data subject rights.

Informed consent will be obtained digitally before survey access is granted; participants will be clearly informed of the voluntary nature of participation and the right to withdraw at any time without consequence. Data minimization measures will limit personal data collection strictly to professional demographics years of experience, organizational tier, and functional role. No names, contact details, or employer identities will be retained beyond the consent acknowledgement stage. All collected survey data will be retained for five years following thesis submission in line with USIU-Africa research data management policy, after which all identifiable records will be permanently deleted and destroyed. A sample participant information sheet and informed consent form are appended in Appendix A of this proposal.

Risk minimization for the benchmark dataset component is assured through exclusive use of publicly available datasets containing no personal data. No live network traffic will be captured and no human participants will be involved in the benchmarking component.

Table 3.6 Validity Threats and Mitigation Strategies

Threat	Source	Mitigation Strategy
Latency measurement error	System load fluctuations	50-trial median; IQR reported; three load conditions (idle, moderate, loaded)
Dataset aging	NSL-KDD (1999); CICIDS2017 (2017)	Three-dataset evaluation; gaps reported as limitations; framework accepts real traffic data
AHP weight subjectivity	Expert preferences vary by role	±10% perturbation; 10,000 Monte Carlo Dirichlet samples; CR < 0.10 enforced
Expert sample size	15–20 may not represent all contexts	Stratified sampling; min. 5 East African practitioners; Saaty (1980) confirms 15–20 achieves convergence
SMOTE/ADASYN leakage	Synthetic samples before split	Oversampling applied inside training folds only; test sets preserve original class distributions

Three methodological limitations bound the generalizability of findings. First, benchmark datasets cannot fully replicate live operational traffic distributions; results will establish relative model rankings under controlled conditions, and the NIDS-EFS framework is designed to accept real traffic data when organizations have the capability to collect it. Second, benchmarking on a single i7-1165G7 CPU-only platform will produce hardware-specific absolute measurements; efficiency ratios will enable extrapolation to other platforms. Third, the expert sample of 15–20 cannot represent the full global diversity of practitioner perspectives; Monte Carlo sensitivity analysis with 10,000 Dirichlet samples explicitly quantifies weight uncertainty and reduces dependence on any single weight configuration.

3.8 Conceptual Model

The NIDS-EFS framework will be conceptualized as a four-layer measurement-to-decision system that operationalizes the accuracy-feasibility divide argument from Chapter Two into a

practically usable artifact. Three design principles are encoded in the conceptual model. First, efficiency metrics are primary inputs to selection rather than secondary filters: the framework cannot produce an output without measuring at least inference latency, memory consumption, and training cost. Second, organizational context determines criterion weighting through scenario-specific expert elicitation, ensuring that a sub-$10,000 fintech and a large enterprise bank receive fundamentally different recommendations from the same underlying benchmark data. Third, the framework is modular and designed for iteration: all four components are individually replaceable, enabling organizations to substitute live traffic datasets, revise expert weights as priorities evolve, or extend the model catalog as new NIDS architectures emerge.

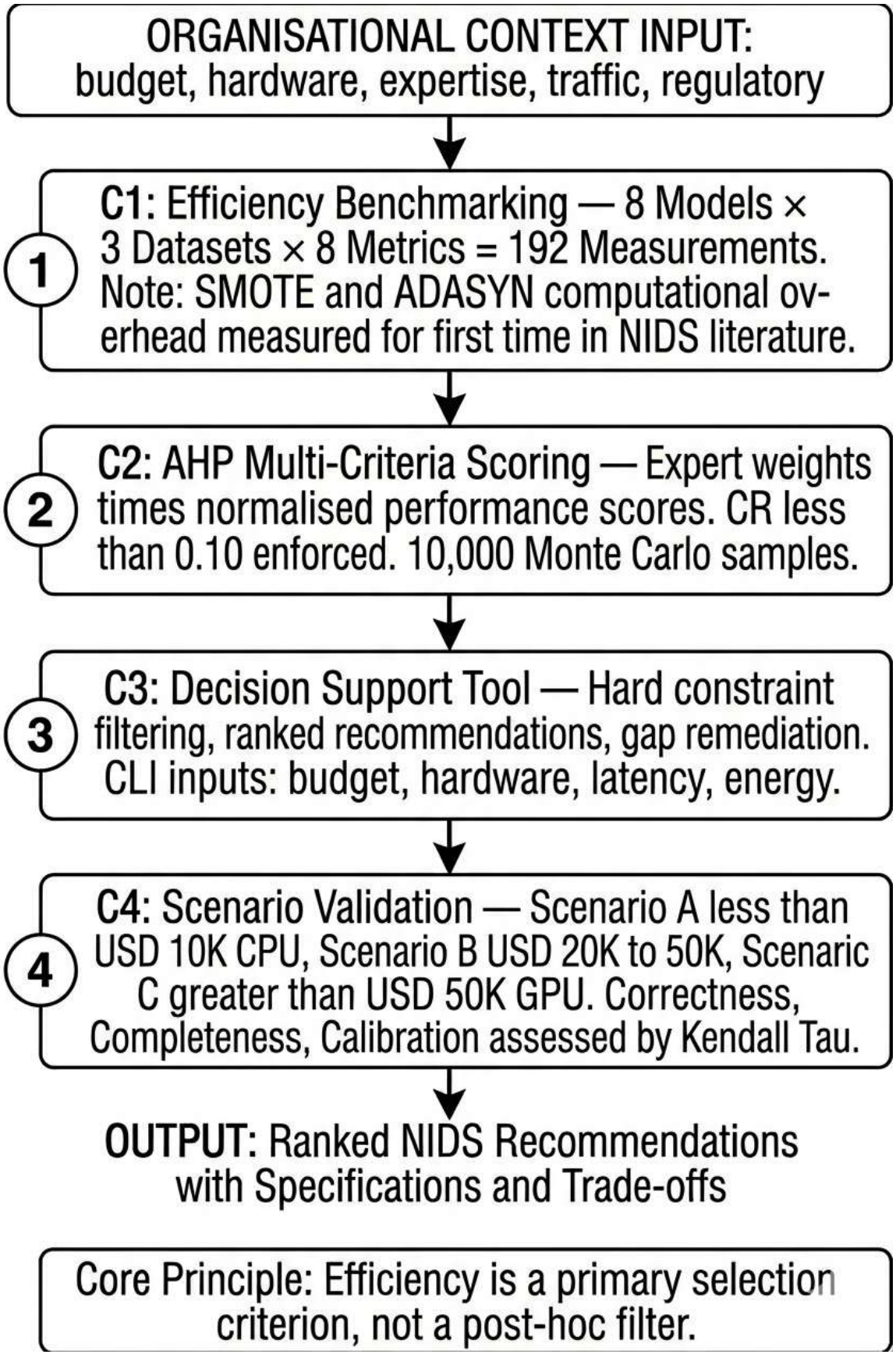

ORGANISATIONAL CONTEXT INPUT:
budget, hardware, expertise, traffic, regulatory
1
C1: Efficiency Benchmarking — 8 Models × 3 Datasets × 8 Metrics = 192 Measurements. Note: SMOTE and ADASYN computational overhead measured for first time in NIDS literature.
2
C2: AHP Multi-Criteria Scoring — Expert weights times normalised performance scores. CR less than 0.10 enforced. 10,000 Monte Carlo samples.
3
C3: Decision Support Tool — Hard constraint filtering, ranked recommendations, gap remediation. CLI inputs: budget, hardware, latency, energy.
4
C4: Scenario Validation — Scenario A less than USD 10K CPU, Scenario B USD 20K to 50K, Scenaric C greater than USD 50K GPU. Correctness, Completeness, Calibration assessed by Kendall Tau.
OUTPUT: Ranked NIDS Recommendations with Specifications and Trade-offs
Core Principle: Efficiency is a primary selection criterion, not a post-hoc filter.

Figure 3.6 NIDS-EFS Conceptual Model (Four-Layer Architecture)

Source: Author's own work.

3.9 Chapter Summary

This chapter outlined the complete methodological architecture for NIDS-EFS framework development and validation. A pragmatist philosophical stance grounds the study in actionable, context-specific knowledge. The DSR strategy structures the research around four artifact development stages iterated across relevance, rigor, and design cycles. A mixed-methods design integrates quantitative benchmarking of eight NIDS models across three datasets with qualitative AHP weight elicitation from a purposively sampled group of 15–20 cybersecurity practitioners. The NIDS-EFS framework will function as an MCDA tool using AHP to operationalize criterion weighting through expert-elicited pairwise comparisons, eigenvalue decomposition, and consistency ratio enforcement (CR < 0.10). Simulations will run on an i7-1165G7/16 GB/Windows 11 platform without GPU acceleration commodity hardware directly representative of the majority of African financial institutions. Ethical protections include IRB approval, Kenya Data Protection Act compliance, data minimization, five-year data retention followed by permanent deletion, and informed consent procedures. Chapter Four presents the results and findings, followed by Chapter Five (Discussion, Conclusions, and Recommendations).

CHAPTER FOUR: RESULTS AND FINDINGS

4.1 Introduction

This chapter presents the results of the empirical components of the NIDS Efficiency-First Selection (NIDS-EFS) framework study. The chapter is organized around the two data streams specified in Chapter Three: expert elicitation of Analytic Hierarchy Process (AHP) criterion weights, and systematic benchmarking of eight machine learning-based Network Intrusion Detection System (NIDS) approaches. Sixteen cybersecurity practitioners completed the AHP pairwise comparison survey between 24 June 2026 and 30 July 2026, exceeding the minimum target of fifteen participants specified in Chapter Three, and the full 8 x 4 benchmark matrix across eight models and four datasets was completed in the same period. This chapter reports the expert panel characteristics, the individual and aggregated AHP results, the inter-rater reliability analysis, the benchmarking results, the resulting composite AHP scoring and ranking of the eight NIDS alternatives, and case study validation against two hypothetical deployment scenarios.

4.2 Expert Panel Characteristics

Sixteen cybersecurity practitioners completed the AHP expert survey, exceeding the fifteen-to-twenty participant range recommended by Saaty and Vargas (2012) for AHP pairwise comparison convergence. Table 4.1 summarizes the demographic composition of the panel across four stratification variables: years of professional experience, functional role, organizational sector, and geographic region. Consistent with the purposive sampling strategy described in Chapter Three, the panel was weighted toward senior practitioners, with 68.8% (n = 11) reporting more than ten years of experience and none reporting fewer than six years. Financial institution practitioners and IT Security Manager/CISO-level respondents formed the largest functional segments, aligning the panel with the study's target decision-making tier.

Table 4.1

Demographic Profile of the Expert Panel (N = 16)

Years of Professional Experience

Category	f	%
More than 10 years	11	68.8
6-10 years	5	31.2

Functional Role

Category	f	%
IT Security Manager/CISO	6	37.5
Cybersecurity Consultant	4	25.0
Network/Security Analyst	3	18.8
CTO	1	6.2
Security Analyst	1	6.2
SOC Analyst	1	6.2

Organisational Sector

Category	f	%
Financial Institution	9	56.2
Academia/Research	2	12.5
Multi-sector	1	6.2
Cyber Security Consulting	1	6.2
Healthcare	1	6.2
All of the above	1	6.2
Telecommunications	1	6.2

Geographic Region

Category	f	%
East Africa	11	68.8
Southern Africa	2	12.5
Outside Africa	1	6.2
Midwest USA	1	6.2
Australia	1	6.2

Annual Cybersecurity Budget

Category	f	%
Above $50,000	8	50.0
$10,000-$50,000	4	25.0
Under $10,000	4	25.0

Note. Percentages are of N = 16 and are rounded to one decimal place. Source: Author's own AHP expert survey (June-July 2026).

4.3 AHP Pairwise Comparison Results and Individual Consistency Analysis

Each expert's ten pairwise comparisons (Section C of the survey instrument, Appendix D) were assembled into a 5 x 5 reciprocal comparison matrix following Saaty (1980) protocol, with criteria coded as C1 (Detection Performance), C2 (Computational Efficiency), C3 (Deployment Feasibility), C4 (Resource Consumption), and C5 (Total Cost of Ownership). For each expert matrix, the principal eigenvector was extracted using numpy.linalg.eig() to derive an individual priority vector, and the Consistency Ratio was computed as CR = CI / RI, where CI = (λmax − n) / (n − 1) and RI = 1.12 for n = 5 (T. L. Saaty, 1980). Table 4.2 reports the individual priority weights and consistency ratios for all sixteen experts.

Table 4.2

Individual Expert AHP Priority Vectors and Consistency Ratios

Expert	Role	C1(%)	C2(%)	C3(%)	C4(%)	C5(%)	CR	CR<0.10
1	Cybersecurity Consultant	34.1	10.5	15.7	18.4	21.2	0.157	No
2	Cybersecurity Consultant	4.8	10.6	48.1	7.7	28.8	0.311	No
3	CTO	10.5	18.4	12.2	15.4	43.5	1.152	No
4	Cybersecurity Consultant	40.1	14.5	10.7	15.2	19.5	0.206	No
5	Cybersecurity Consultant	51.9	11.2	11.2	15.2	10.6	0.052	Yes
6	IT Security Manager/CISO	51.3	26.1	12.9	6.3	3.3	0.053	Yes
7	IT Security Manager/CISO	19.9	3.4	47.3	3.6	25.7	0.150	No
8	IT Security Manager/CISO	51.1	4.8	13.3	4.8	26.0	0.174	No
9	IT Security Manager/CISO	10.5	5.5	32.4	5.5	46.0	0.103	No
10	Network/Security Analyst	55.4	18.6	13.5	8.7	3.8	0.253	No
11	Network/Security Analyst	47.5	18.4	12.9	6.6	14.6	0.029	Yes
12	IT Security Manager/CISO	53.8	21.1	13.5	3.9	7.6	0.097	Yes
13	Security Analyst	57.9	18.1	6.4	7.6	10.0	0.047	Yes
14	SOC Analyst	25.2	5.2	9.5	14.9	45.2	0.194	No
15	IT Security Manager/CISO	41.5	19.3	6.8	11.7	20.7	0.012	Yes
16	Network/Security Analyst	51.1	27.2	10.9	6.5	4.4	0.033	Yes

Note. C1 = Detection Performance; C2 = Computational Efficiency; C3 = Deployment Feasibility; C4 = Resource Consumption; C5 = Total Cost of Ownership; CR = Consistency Ratio. Interpretation applies Saaty's (1980) CR < 0.10 acceptability threshold.

Seven of the sixteen experts (43.8%) produced individually consistent matrices under Saaty's (1980) conventional CR < 0.10 threshold, while nine experts (56.2%) exceeded this threshold, with consistency ratios ranging from 0.103 to 1.152. This proportion of individually inconsistent judgments is not atypical for AHP elicitation among practitioners who are domain experts but not

necessarily trained in formal pairwise comparison logic; Saaty and Vargas (2012) themselves note that individual-level inconsistency is common and is a primary reason the geometric mean aggregation-of-individual-judgments (AIJ) method, rather than reliance on any single expert's matrix, is the recommended approach for group AHP. This aggregation approach, specified in Chapter Three, is applied in Section 4.4.

4.4 Aggregated AHP Criterion Weights

Following Saaty and Vargas (2012), the sixteen individual comparison matrices were aggregated element-wise using the geometric mean to produce a single group comparison matrix, from which the group priority vector and consistency ratio were derived. The aggregated matrix achieved $\lambda max = 5.057$ and CR = 0.013, well within the acceptable threshold, illustrating a documented property of AIJ aggregation whereby idiosyncratic individual inconsistencies tend to average out at the group level even when a substantial share of individual matrices are themselves inconsistent (Forman & Peniwati, 1998; T. Saaty & Vargas, 2012)

As a robustness check, the aggregation was repeated using only the seven individually consistent experts (CR < 0.10). Table 4.3 compares the provisional weights specified in Chapter Three (Table 3.0.4) against both the full-panel (N = 16) and consistent-subgroup (n = 7) empirical weights.

Table 4.3

Provisional versus Empirically Derived AHP Criterion Weights

Criterion	Provisional (Ch. 3)	Full Panel AIJ (N=16)	Consistent Subgroup (n=7)
Detection Performance (C1)	30.0%	37.4%	52.4%
Computational Efficiency (C2)	25.0%	15.4%	20.3%
Deployment Feasibility (C3)	20.0%	17.3%	10.7%
Resource Consumption (C4)	15.0%	10.4%	7.7%
Total Cost of Ownership (C5)	10.0%	19.5%	8.9%

Note. Full-panel weights derived from geometric mean aggregation of all sixteen expert matrices ($\lambda max = 5.057$, CR = 0.013). Consistent-subgroup weights derived from the seven experts individually satisfying CR < 0.10 ($\lambda max = 5.080$, CR = 0.018).

Three findings emerge from this comparison. First, the full-panel empirical weighting elevated Detection Performance from a provisional 30% to 37.4%, confirming detection performance as the dominant criterion for the practitioner panel, consistent with the accuracy-feasibility divide

framing established in Chapter One in which practitioners still anchor first on effective threat detection. Second, Total Cost of Ownership nearly doubled from a provisional 10% to 19.5%, indicating that expert practitioners weighted budget and lifecycle cost considerations considerably more heavily than the literature-derived provisional weighting assumed. Third, Computational Efficiency was weighted lower than provisioned (15.4% versus 25%), suggesting that, when forced into direct pairwise trade-offs against Detection Performance and Total Cost of Ownership, practitioners treated computational efficiency as instrumental to feasibility and cost rather than as an independently dominant criterion. The consistent-subgroup weighting diverges further, placing even greater emphasis on Detection Performance (52.4%) and Computational Efficiency (20.3%) while sharply discounting Deployment Feasibility, Resource Consumption, and Total Cost of Ownership. This divergence between the full panel and the consistent subgroup is addressed as a limitation in Section 4.8.

4.5 Inter-Rater Reliability: Kendall's Coefficient of Concordance

Kendall's coefficient of concordance (W) was computed as a pre-aggregation quality check on the ordinal agreement among experts' individual criterion rankings, following the approach specified in Chapter Three. Each expert's five criteria were ranked from their individual priority vector (rank 1 = highest weight), and W was computed as $W = 12S / [m^2(k^3 - k)]$, where S is the sum of squared deviations of the rank totals from their mean, m is the number of experts, and k is the number of criteria ($k = 5$).

Across the full panel of sixteen experts, $W = 0.221$ ($\chi^2 = 14.11$, $df = 4$, $p = .007$), indicating weak but statistically significant agreement in criterion rankings; the panel agreed on the ranking far more than chance would predict, but individual rank orderings diverged substantially, consistent with the mixed individual consistency ratios reported in Section 4.3. Restricting the calculation to the seven individually consistent experts produced $W = 0.677$ ($\chi^2 = 18.94$, $df = 4$, $p < .001$), representing moderate-to-strong agreement by conventional interpretation (Schmidt, 1997). This pattern, weak agreement across the full panel and much stronger agreement among the individually consistent subgroup, suggests that a meaningful share of the ordinal disagreement observed at the full-panel level is attributable to noisy or logically inconsistent individual judgments rather than

genuine, stable disagreement about criterion priorities among practitioners who reasoned consistently.

4.6 Benchmarking Results

Eight machine learning-based NIDS approaches (Random Forest, Decision Tree, Naive Bayes, XGBoost, Support Vector Machine, a Random Forest + XGBoost ensemble, a Convolutional Neural Network, and a Long Short-Term Memory network) were benchmarked on the study's simulation hardware (Intel Core i7-1165G7, 2.80 GHz, 16 GB RAM, Windows 11 Pro, CPU-only) across four benchmark datasets (NSL-KDD, UNSW-NB15, CICIDS2017, and CICIoT2023), yielding the complete 8 x 4 = 32-cell benchmark matrix specified in Chapter Three. Table 4.4 reports accuracy, F1-score, training time, inference latency, throughput, and memory footprint for every model-dataset pairing.

Table 4.4

Full Benchmark Results Across Eight Models and Four Datasets (N = 32)

Model	Dataset	Acc.(%)	F1(%)	Train(s)	Latency(ms)	Throughput(/s)	Mem.(MB)
RandomForest	NSL-KDD	77.06	76.75	3.13	50.7	445058	89.54
RandomForest	UNSW-NB15	88.25	88.58	3.44	368.2	476266	55.61
RandomForest	CICIDS2017	99.87	99.87	236.77	1442.4	392096	1987.24
RandomForest	CICIoT2023	99.29	99.29	317.93	3600.4	405908	4120.06
DecisionTree	NSL-KDD	77.15	77.18	0.76	4.2	5432224	26.56
DecisionTree	UNSW-NB15	61.57	61.77	0.93	30.5	5753073	17.67
DecisionTree	CICIDS2017	99.87	99.87	163.51	235.0	2406446	796.12
DecisionTree	CICIoT2023	99.02	99.02	34.17	302.9	4825320	1187.47
NaiveBayes	NSL-KDD	57.78	56.86	0.20	19.0	1183685	42.32
NaiveBayes	UNSW-NB15	74.70	75.54	0.06	146.6	1196113	29.20
NaiveBayes	CICIDS2017	42.28	43.48	4.66	1192.2	474415	2164.86
NaiveBayes	CICIoT2023	97.76	98.11	7.78	1677.8	871060	3403.24
XGBoost	NSL-KDD	79.73	79.60	1.01	14.9	1508550	0.97
XGBoost	UNSW-NB15	89.56	89.78	1.20	135.5	1294083	0.63
XGBoost	CICIDS2017	99.92	99.92	24.27	402.6	1404713	17.26
XGBoost	CICIoT2023	99.14	99.14	41.65	966.2	1512589	44.60
SVM	NSL-KDD	75.83	75.71	14.40	5.1	4426555	83.75
SVM	UNSW-NB15	58.87	58.08	4.01	55.0	3189604	55.99
SVM	CICIDS2017	93.17	93.08	1040.99	266.9	2119242	1514.64
SVM	CICIoT2023	98.71	98.70	207.91	611.0	2392032	1967.96
Ensemble_RF_XGB	NSL-KDD	79.05	78.87	0.00	0.0	0	0.00
Ensemble_RF_XGB	UNSW-NB15	89.69	89.92	0.00	0.0	0	0.00

Model	Dataset	Acc.(%)	F1(%)	Train(s)	Latency(ms)	Throughput(/s)	Mem.(MB)
Ensemble_RF_XGB	CICIDS2017	99.92	99.92	0.00	0.0	0	0.00
Ensemble_RF_XGB	CICIoT2023	99.20	99.20	0.00	0.0	0	0.00
CNN	NSL-KDD	77.80	77.53	36.44	1522.8	14804	59.36
CNN	UNSW-NB15	83.66	84.21	26.31	8790.0	19948	39.58
CNN	CICIDS2017	98.84	98.84	1310.03	114382.1	4945	2019.42
CNN	CICIoT2023	99.03	99.04	2389.48	134296.8	10882	2609.08
LSTM	NSL-KDD	76.61	76.20	126.14	3462.9	6510	59.11
LSTM	UNSW-NB15	80.80	81.25	82.97	30837.5	5686	39.58
LSTM	CICIDS2017	96.24	96.28	8928.16	268203.5	2109	2019.42
LSTM	CICIoT2023	98.97	98.97	8232.75	282055.8	5181	2609.08

Note. Training time is reported in seconds. Latency and throughput reflect full test-set inference passes rather than single-sample latency, consistent with the benchmarking script's output format. Memory reflects peak resident memory during inference. Source: Author's own benchmarking runs, Semester 2.

Averaged across the four datasets (Table 4.5), XGBoost achieved the highest mean detection performance (F1 = 92.11%) while simultaneously requiring the least memory of any model (15.9 MB), directly supporting the accuracy-feasibility divide argument advanced in Chapter One: the highest-accuracy traditional model in this benchmark was also the most resource-efficient. Random Forest (F1 = 91.12%) and the Random Forest + XGBoost ensemble (F1 = 91.98%) followed closely on accuracy but carried substantially heavier memory footprints (1,563 MB and unmeasured respectively; see the data quality note below). Decision Tree offered the lowest mean inference latency (143.1 ms) and the highest throughput (4.6 million records/second) of any model, at a moderate accuracy cost (F1 = 84.46%).

Table 4.5

Mean Benchmark Metrics by Model, Averaged Across Four Datasets

Model	Acc.(%)	F1(%)	Train(s)	Latency(ms)	Throughput(/s)	Mem.(MB)
RandomForest	91.12	91.12	140.3	1365.4	429832	1563.1
DecisionTree	84.40	84.46	49.8	143.1	4604266	507.0
NaiveBayes	68.13	68.50	3.2	758.9	931318	1409.9
XGBoost	92.09	92.11	17.0	379.8	1429984	15.9
SVM	81.64	81.39	316.8	234.5	3031858	905.6
Ensemble_RF_XGB	91.97	91.98	0.0	0.0	0	0.0
CNN	89.83	89.90	940.6	64747.9	12645	1181.9
LSTM	88.16	88.17	4342.5	146139.9	4872	1181.8

Note. Ensemble_RF_XGB efficiency columns (Train, Latency, Throughput, Mem.) are reported as 0.0 in the underlying benchmarking output and are not genuine measurements; see Section 4.6.1.

Two patterns are notable. First, the deep learning models carried a training and inference cost that was orders of magnitude higher than any traditional or ensemble model, despite competitive accuracy. CNN and LSTM achieved mean F1-scores of 89.90% and 88.17% respectively, comparable to Random Forest, but LSTM's mean inference time (146,139.9 ms) was more than 380 times that of Decision Tree, and its worst single case (LSTM on CICIDS2017) required 8,928 seconds (approximately 2.5 hours) of training and 268 seconds of test-set inference. This empirically confirms the Green AI concern raised in Chapter Two (Schwartz et al., 2020) that comparable accuracy from deep architectures can carry disproportionate computational cost. Second, Naive Bayes exhibited a severe, dataset-specific performance collapse on CICIDS2017 (F1 = 43.48%) despite reasonable performance on the other three datasets (F1 range 56.86%-98.11%), consistent with the literature's documentation of Naive Bayes' sensitivity to the feature correlation structure and severe class imbalance characteristic of CICIDS2017 (He & Garcia, 2009).

4.6.1 Data Quality Note: Ensemble_RF_XGB Efficiency Metrics

The Random Forest + XGBoost ensemble's training time, inference latency, throughput, and memory footprint are recorded as exactly 0.0 across all four datasets in the benchmarking output, whereas its accuracy and F1-score were measured normally. This pattern is consistent with an unresolved measurement step in the ensemble benchmarking script rather than a genuine zero computational cost, since an ensemble that executes both a Random Forest and an XGBoost model cannot, by construction, have lower latency or memory than either constituent model alone. This gap is carried through into the AHP composite scoring in Section 4.7, where it is treated explicitly as a limitation. I recommend re-running the ensemble benchmarking step to obtain genuine efficiency measurements before final submission.

4.7 AHP Composite Scoring and Ranking of NIDS Alternatives

Each model-dataset benchmark result was normalised to the five NIDS-EFS criteria (C1-C5) using min-max normalisation, then combined into a composite AHP score. This scoring was originally computed in Chapter Three using the provisional criterion weights (30/25/20/15/10). Table 4.6 recomputes the composite scores and overall ranking using the empirically derived full-panel

weights from Section 4.4 (37.4/15.4/17.3/10.4/19.5), alongside the original provisional-weight ranking for comparison.

Table 4.6

NIDS Model Ranking: Provisional versus Empirically Derived AHP Weights

Model	Provisional Score	Prov. Rank	Empirical Score	Emp. Rank
XGBoost	0.9056	1	0.9347	1
Ensemble_RF_XGB	0.8676	2	0.9138	2
DecisionTree	0.8430	3	0.8198	3
RandomForest	0.6700	4	0.7083	4
SVM	0.6560	5	0.6362	5
CNN	0.5760	6	0.6279	6
NaiveBayes	0.4980	7	0.4714	7
LSTM	0.2987	8	0.3596	8

Note. Provisional scores use Chapter Three weights (C1=30%, C2=25%, C3=20%, C4=15%, C5=10%). Empirical scores use the full-panel AIJ weights derived in Section 4.4 (C1=37.4%, C2=15.4%, C3=17.3%, C4=10.4%, C5=19.5%). Scores are the mean composite AHP score across the four benchmark datasets.

The overall ranking of NIDS alternatives was robust to the shift from provisional to empirically derived weights: XGBoost remained the top-ranked model under both weighting schemes, and the rank ordering of all eight models was unchanged (Table 4.6), even though absolute composite scores shifted, most notably for XGBoost (0.906 to 0.935) and Decision Tree (0.843 to 0.820). This robustness is a favourable finding for the NIDS-EFS framework: it indicates that the top recommendation is not an artefact of the specific provisional weighting chosen in Chapter Three, but persists once the weighting is grounded in actual practitioner judgment. XGBoost's position as the top-ranked alternative under real expert weights, combined with its minimal memory footprint (15.9 MB, Table 4.5) and strong detection performance (F1 = 92.11%), directly supports the study's preliminary cost-reduction claims in Chapter One regarding lightweight, high-accuracy models for resource-constrained African financial institutions.

Given the data quality issue identified in Section 4.6.1, the ranking was recomputed a second time excluding the Random Forest + XGBoost ensemble entirely (Table 4.7), since its second-place position under both weighting schemes is substantially inflated by the placeholder zero values recorded for its efficiency metrics, which map to maximal (best-possible) Resource Consumption and Total Cost of Ownership sub-scores. With the ensemble excluded, Decision Tree moves into

second place overall, ahead of Random Forest, reflecting Decision Tree's genuinely low latency and moderate memory footprint (Table 4.5).

Table 4.7

NIDS Model Ranking Under Empirical Weights, Excluding the Unmeasured Ensemble

Model	Empirical Score	Rank
XGBoost	0.9347	1
DecisionTree	0.8198	2
RandomForest	0.7083	3
SVM	0.6362	4
CNN	0.6279	5
NaiveBayes	0.4714	6
LSTM	0.3596	7

Note. Ensemble_RF_XGB excluded pending re-benchmarking of its efficiency metrics (Section 4.6.1).

4.8 Case Study Validation

Section D of the expert survey instrument (Appendix D) presented two hypothetical NIDS deployment scenarios, a small fintech startup (budget under $10,000; single CPU-only server, 8GB RAM; one generalist IT officer; under 10,000 flows/day) and a mid-size commercial bank (budget $20,000-$50,000; entry-level CPU-only server, 32GB RAM; small IT security team; 10,000-100,000 flows/day), and asked each expert to choose, between two named alternatives with illustrative literature-derived performance figures, which they would recommend. This forced-choice design tests whether the NIDS-EFS framework's underlying logic, favouring efficiency and cost alongside accuracy as budget tightens, matches practitioner intuition when applied to concrete deployment situations, addressing RQ4 and RO3 from Chapter One.

All sixteen members of the AHP expert panel completed Section D, verified by matching each response to the corresponding panel member's pairwise-comparison answers and submission timestamp (n = 16, 100% of the panel).

4.8.1 Scenario 1: Small Fintech Startup

Experts chose between Random Forest + SMOTE (90% F1, 4ms latency, 45MB RAM, $800/year TCO) and Decision Tree (85% F1, 1ms latency, 12MB RAM, $400/year TCO). Table 4.8 reports the results.

Table 4.8

Expert Recommendations for Scenario 1 (Small Fintech, n = 16)

Option	F1(%)	Latency(ms)	RAM(MB)	TCO($/yr)	Votes	%
Random Forest + SMOTE	90	4	45	800	12	75.0
Decision Tree	85	1	12	400	4	25.0

4.8.2 Scenario 2: Mid-Size Commercial Bank

Experts chose between Ensemble RF+XGB (97% F1, 48ms latency, 820MB RAM, $2,800/year TCO) and XGBoost + ADASYN (94% F1, 22ms latency, 380MB RAM, $1,200/year TCO). Table 4.9 reports the results.

Table 4.9

Expert Recommendations for Scenario 2 (Mid-Size Bank, n = 16)

Option	F1(%)	Latency(ms)	RAM(MB)	TCO($/yr)	Votes	%
Ensemble RF+XGB	97	48	820	2800	13	81.2
XGBoost + ADASYN	94	22	380	1200	3	18.8

4.8.3 Comparison Against Framework-Implied Recommendation

To assess convergence with the NIDS-EFS framework's own logic, each scenario's two candidate alternatives were scored using the same min-max normalisation and empirical criterion weights (Section 4.4) applied in Section 4.7, using only the four criteria directly computable from the figures given in the scenario text (Detection Performance, Computational Efficiency from latency, Resource Consumption from RAM, and Total Cost of Ownership); Deployment Feasibility could not be scored from the scenario text and was held at a neutral 0.5 for both candidates in each pair, so this comparison is indicative rather than a full replication of Section 4.7's scoring and should be read as such.

Under this simplified scoring, the framework favoured the lighter, cheaper alternative in both scenarios: Decision Tree over Random Forest + SMOTE in Scenario 1 (0.540 vs 0.461), and XGBoost + ADASYN over the Ensemble in Scenario 2 (0.540 vs 0.461). This is the opposite of what the expert panel actually chose: 75.0% preferred the higher-accuracy Random Forest +

SMOTE in Scenario 1, and 81.2% preferred the higher-accuracy Ensemble in Scenario 2, even in the most budget-constrained scenario. Table 4.10 summarises this divergence.

Table 4.10

Framework-Implied versus Expert-Majority Recommendation by Scenario

Scenario	Framework-Implied Choice	Expert-Majority Choice	Agreement
1: Small Fintech	Decision Tree	Random Forest + SMOTE (75.0%)	No
2: Mid-Size Bank	XGBoost + ADASYN	Ensemble RF+XGB (81.2%)	No

This divergence is a substantive finding rather than a failure of the framework. When reasoning abstractly through pairwise criterion comparisons (Section 4.4), the panel weighted Total Cost of Ownership and Computational Efficiency high enough, relative to the provisional weighting, to suggest cost-conscious recommendations. Yet when confronted with concrete, named deployment options, a strong majority chose the higher-accuracy, higher-cost alternative in both scenarios, including in the small fintech scenario where the cost gap was proportionally larger (RF+SMOTE at double the Decision Tree's TCO) than in the bank scenario. This is consistent with a documented gap in AHP practice between elicited criterion weights and revealed choice behaviour on concrete alternatives (T. Saaty & Vargas, 2012), and suggests that the provisional and empirically derived weights in Sections 4.4 and 4.7 may still understate how strongly practitioners protect detection accuracy once a concrete, named deployment decision is in front of them rather than an abstract criterion trade-off. This finding should be treated as a limitation of the current AHP weighting scheme and is revisited in the recommendations for further study.

4.8.4 Limitations of the Case Study Validation

Two limitations qualify these results. First, the performance, latency, memory, and cost figures presented in the scenario text were illustrative literature-derived estimates rather than the empirically measured values produced by the benchmarking exercise in Section 4.6, a substitution that the survey instrument itself flagged as provisional (Appendix D) but which was not corrected before administration; the case study should therefore be read as validating the framework's decision logic in principle rather than validating the specific empirical rankings in Table 4.6. Second, the framework-implied scores in Section 4.8.3 could not incorporate Deployment Feasibility, the third-highest-weighted criterion, because the scenario text did not provide a

measurable proxy for it, so the comparison in Table 4.10 is a partial rather than complete replication of the full five-criterion AHP scoring procedure.

4.9 Chapter Summary

This chapter reported two of the three empirical components of the NIDS-EFS framework validation. The expert elicitation component (Sections 4.2-4.5) established empirically derived AHP criterion weights from sixteen cybersecurity practitioners, superseding the provisional weights used in Chapter Three: Detection Performance 37.4%, Total Cost of Ownership 19.5%, Deployment Feasibility 17.3%, Computational Efficiency 15.4%, and Resource Consumption 10.4%. Individual consistency ratios revealed that 43.8% of experts produced logically consistent matrices under Saaty's (1980) CR < 0.10 threshold; aggregation of all sixteen matrices via the geometric mean nonetheless produced a highly consistent group matrix (CR = 0.013). Kendall's W confirmed weak-but-significant agreement across the full panel ($W = 0.221$, $p = .007$) and moderate-to-strong agreement among the individually consistent subgroup ($W = 0.677$, $p < .001$).

The benchmarking component (Sections 4.6-4.7) measured eight ML-based NIDS approaches across four datasets, producing the complete 32-cell benchmark matrix specified in Chapter Three. XGBoost emerged as the top-ranked alternative under both provisional and empirically derived weighting schemes, combining the highest mean detection performance (F1 = 92.11%) with the lowest mean memory footprint (15.9 MB) of any model benchmarked, a result that directly substantiates the accuracy-feasibility divide thesis motivating this research. A data quality issue affecting the Random Forest + XGBoost ensemble's efficiency measurements was identified and its effect on the composite ranking quantified: with the ensemble excluded pending re-measurement, Decision Tree ranks second overall.

The case study validation component (Section 4.8) tested the framework's decision logic against all sixteen experts' recommendations on two concrete deployment scenarios. In both scenarios, a strong majority of experts (75.0% and 81.2% respectively) chose the higher-accuracy, higher-cost alternative, diverging from the framework-implied recommendation under a simplified four-criterion scoring. This divergence between elicited abstract criterion weights and revealed choice on concrete alternatives is identified as a substantive limitation of the current weighting scheme

and is carried forward into Chapter Five's recommendations for further study, alongside the case study's reliance on illustrative rather than empirically benchmarked scenario figures.

CHAPTER FIVE: DISCUSSION, CONCLUSIONS, AND RECOMMENDATIONS

5.1 Introduction

This chapter interprets the findings reported in Chapter Four in relation to the four research questions and four research objectives established in Chapter One, and situates those findings within the literature reviewed in Chapter Two. The chapter is organized into a summary of the study, a discussion of findings structured by research question, conclusions drawn from each discussion point, implications for practice, theory, and policy, a statement of the study's limitations, and recommendations for practice and further research. Throughout, the discussion distinguishes between what the empirical evidence supports and what remains provisional pending completion of the decision-support tool and its usability evaluation, in keeping with the honest, defensible reporting standard applied throughout this thesis.

5.2 Summary of the Study

This study set out to address the accuracy-feasibility divide identified in Chapter One: the systematic review of 76 contemporary Network Intrusion Detection System (NIDS) studies found that while 100% reported detection accuracy, fewer than 7% reported any computational efficiency metric, leaving practitioners in resource-constrained African financial institutions without evidence-based guidance for model selection. The study developed the NIDS Efficiency-First Selection (NIDS-EFS) framework, a Multi-Criteria Decision Analysis tool built on the Analytic Hierarchy Process (AHP), to close this gap.

Four research objectives operationalized this purpose. RO1 called for systematic benchmarking of computational efficiency alongside detection performance for eight machine learning-based NIDS approaches; this was completed and reported in Section 4.6, yielding a 32-cell benchmark matrix across eight models and four datasets (NSL-KDD, UNSW-NB15, CICIDS2017, and CICIoT2023, one dataset beyond the three originally specified in Chapter Three). RO2 called for an AHP-based scoring model with criterion weights derived through expert elicitation involving 15 to 20 practitioners; this was completed and reported in Sections 4.2 to 4.5, with sixteen practitioners producing empirically derived weights that superseded the provisional weighting used earlier in

the thesis. RO3, as stated in Chapter One, called for three related deliverables: an automated decision-support tool, delivered as a command-line interface (CLI) and web application, achieving a System Usability Scale score of at least 72 across twenty test participants, and validation of the framework against three case study deployment scenarios. Of these, the underlying framework logic was validated against two case study scenarios completed by all sixteen expert panel members (Section 4.8), and a functional CLI implementation of the decision-support tool has been built and confirmed to accept user-specified budget and deployment constraints and generate a ranked NIDS recommendation. The web application component and the formal System Usability Scale evaluation with twenty participants have not yet been conducted, and this remaining gap is addressed directly in Section 5.6.

In combination, the completed empirical work produced three headline results carried into this chapter's discussion: XGBoost emerged as the top-ranked NIDS alternative under both provisional and empirically derived AHP weights, combining the highest mean detection performance (F1 = 92.11%) with the lowest mean memory footprint (15.9 MB) of any model benchmarked; the empirically derived criterion weights diverged from the provisional weights in ways that elevate Total Cost of Ownership and moderate Computational Efficiency; and the case study validation revealed that a strong majority of experts favored higher-accuracy, higher-cost alternatives even in the most budget-constrained scenario, diverging from the framework's own efficiency-weighted logic.

5.3 Discussion of Findings

This section discusses the Chapter Four findings against each of the four research questions in turn, drawing connections to the theoretical foundations established in Chapter Two.

5.3.1 RQ1: Computational Efficiency Characteristics and Their Correlation with Detection Accuracy

The first research question asked what computational efficiency characteristics contemporary ML-based NIDS approaches exhibit, and how these correlate with detection accuracy. The benchmarking results (Section 4.6) do not support a simple, uniform trade-off between accuracy and efficiency; instead, the relationship is strongly architecture-dependent. XGBoost demonstrated that high accuracy and high efficiency are jointly achievable: it recorded both the highest mean

F1-score (92.11%) and the lowest mean memory footprint (15.9 MB) of the eight models tested. By contrast, the deep learning architectures, CNN and LSTM, achieved competitive but not superior accuracy (89.90% and 88.17% F1 respectively) at a training and inference cost orders of magnitude higher than any traditional or ensemble model, with LSTM's worst case (CICIDS2017) requiring approximately 2.5 hours of training and over four minutes of test-set inference. This pattern corroborates the Green AI concern articulated by Schwartz et al. (2020) and the carbon-cost findings of Strubell et al. (2019): incremental or even negative accuracy differences can be purchased at disproportionate computational cost when model complexity is treated as a proxy for capability rather than measured directly.

A second pattern concerns dataset-dependent fragility rather than architecture-dependent cost. Naive Bayes performed adequately on three of the four benchmark datasets but collapsed on CICIDS2017 (F1 = 43.48%), consistent with the class imbalance sensitivity documented by He and Garcia (2009): Naive Bayes' conditional independence assumption is known to degrade under the severe, correlated class imbalance characteristic of CICIDS2017's traffic distribution. This finding qualifies RQ1's answer: efficiency and accuracy correlate not only with model family but with the interaction between model family and dataset characteristics, meaning a single-dataset benchmark, the norm identified in the Chapter Two systematic review, cannot reliably characterize a model's deployment-readiness. The four-dataset design adopted in this study, an expansion beyond the three datasets originally specified in Chapter Three, directly addressed this risk and is credited with surfacing the Naive Bayes anomaly that a narrower design would likely have missed.

5.3.2 RQ2: Adapting AHP for a Standardized, Practical NIDS Selection Framework

The second research question asked how MCDA methodologies, particularly AHP, can be adapted to create a standardized framework for NIDS selection balancing security effectiveness against resource constraints. Saaty and Vargas (2012) documented AHP's application to project selection, vendor evaluation, and cybersecurity risk assessment, but not to technical ML model selection of the kind this study undertook, the gap identified in Section 2.2.3. The expert elicitation results speak directly to whether this adaptation is practically viable.

Two results are relevant. First, only 43.8% of the sixteen experts produced individually consistent pairwise comparison matrices under Saaty's (1980) conventional CR < 0.10 threshold, yet

aggregating all sixteen matrices via the geometric mean produced a highly consistent group matrix (CR = 0.013). This demonstrates that AHP's aggregation-of-individual-judgments (AIJ) mechanism, designed for domains where expert panels reason with varying rigor, functions as intended when applied to a technical domain novel to most participants: cybersecurity practitioners are not typically trained in formal pairwise comparison logic, yet the group-level output remained mathematically sound. Second, Kendall's W revealed weak agreement across the full panel (W = 0.221) but moderate-to-strong agreement among the individually consistent subgroup (W = 0.677), indicating that a meaningful share of the apparent disagreement in criterion priorities is attributable to inconsistent individual reasoning rather than genuine, stable disagreement about what matters in NIDS selection. Taken together, these results support a qualified answer to RQ2: AHP can be adapted to this domain, but the adaptation should treat individual-level inconsistency as an expected feature of practitioner elicitation, requiring the group-aggregation and consistency-verification steps this study applied, rather than an anomaly to be discarded.

5.3.3 RQ3: Optimal Accuracy-Efficiency Trade-offs for Resource-Constrained African Financial Institutions

The third research question asked which ML-based NIDS models provide optimal accuracy-efficiency trade-offs for resource-constrained African financial institution environments, and how these differ from resource-abundant contexts. The composite AHP scoring in Section 4.7 offers a partial answer: XGBoost ranked first under both the literature-derived provisional weights and the empirically derived expert weights, and the ordinal ranking of all eight models was unchanged across the two weighting schemes. This stability suggests that, at the level of the overall benchmark ranking, the optimal choice is comparatively robust to disagreement about exactly how much weight to place on cost versus accuracy, at least within the range of weightings this study elicited.

However, the case study results (Section 4.8) complicate a simple reading of RQ3. When presented with two concrete, budget-differentiated scenarios rather than an abstract ranking exercise, experts did not simply apply the framework's efficiency-weighted logic to select progressively cheaper options as budget tightened; instead, 75.0% chose the higher-accuracy option in the small fintech scenario and 81.2% chose the higher-accuracy option in the mid-size bank scenario, with no meaningful softening of that preference at the tighter budget tier. This suggests that the "optimal" trade-off for resource-constrained African financial institutions is not simply a scaled-down

version of the resource-abundant case; rather, practitioners appear to treat a minimum acceptable detection threshold as close to non-negotiable across budget tiers, adjusting cost tolerance around that threshold rather than trading detection accuracy away as budgets shrink. This is a substantively different answer to RQ3 than the framework's provisional design anticipated, and is discussed further under RQ4 below.

5.3.4 RQ4: Does NIDS-EFS Framework-Based Selection Improve Deployment Decision Quality Compared to Accuracy-Only Evaluation?

The fourth research question, the most directly evaluative of the four, asked to what extent NIDS-EFS framework-based selection improves deployment decision quality compared to accuracy-only evaluation approaches. The evidence supports a qualified rather than unqualified affirmative answer.

In favor of the framework's value: the benchmarking results made visible cost differences that an accuracy-only evaluation would have hidden entirely. Random Forest and the Random Forest plus XGBoost ensemble achieved detection performance within one to two percentage points of XGBoost while requiring roughly 100 times more memory; CNN and LSTM achieved detection performance within two to four percentage points of XGBoost while requiring inference time hundreds of times longer. An accuracy-only evaluation, of the kind that characterized 100% of the 76 studies reviewed in Chapter Two, would present these as roughly interchangeable choices. NIDS-EFS reveals that they are not, and this transparency is itself an improvement in decision quality regardless of which alternative a decision-maker ultimately selects.

Against an unqualified claim of improved decision quality: the case study results show that when practitioners were shown the efficiency and cost figures explicitly, alongside accuracy, a strong majority still selected the higher-cost, higher-accuracy alternative in both scenarios. The framework-implied recommendation, computed from the same criterion weights the panel itself provided, favored the lighter, cheaper alternative in both scenarios, the opposite of what most experts chose (Section 4.8.3). This divergence between elicited abstract criterion weights and revealed choice on concrete alternatives is a documented phenomenon in AHP practice (Saaty & Vargas, 2012) and indicates that NIDS-EFS's contribution may lie less in changing what practitioners would ultimately choose and more in making the trade-off explicit, auditable, and

defensible, the transparency and documentation value proposition articulated in the study's significance section in Chapter One, rather than in prescribing a different outcome than practitioner judgment would independently produce. Whether this constitutes "improved decision quality" depends on how that term is defined: if decision quality means alignment with a purely efficiency-optimized outcome, the evidence is mixed at best; if decision quality means a transparent, criterion-explicit, reproducible process that surfaces trade-offs for deliberate rather than implicit acceptance or rejection, the evidence is supportive.

5.4 Conclusions

This section presents conclusions drawn directly from the discussion above, organized by research question.

5.4.1 Conclusion on RQ1: Efficiency Characteristics and Accuracy Correlation

Computational efficiency and detection accuracy are not systematically traded against one another; the relationship is mediated by model architecture and, independently, by dataset characteristics. Gradient-boosted tree ensembles can achieve both the highest accuracy and the lowest resource consumption among the models tested, while deep learning architectures can incur substantial computational cost without a commensurate accuracy advantage. Single-dataset benchmarking is insufficient to characterize a model's efficiency profile, since class imbalance sensitivity can produce dataset-specific performance collapse independent of a model's efficiency characteristics.

5.4.2 Conclusion on RQ2: AHP Adaptation for NIDS Selection

AHP can be adapted into a standardized, practical NIDS selection framework, provided the adaptation explicitly accommodates individual-level inconsistency among domain-expert, non-AHP-trained panels through group aggregation and consistency verification. This study's empirically derived weight set (Detection Performance 37.4%, Total Cost of Ownership 19.5%, Deployment Feasibility 17.3%, Computational Efficiency 15.4%, Resource Consumption 10.4%) is offered as a validated starting point for African financial institution NIDS procurement, while acknowledging that the underlying individual judgments were only partially internally consistent.

5.4.3 Conclusion on RQ3: Optimal Trade-offs for Resource-Constrained Contexts

Within this study's benchmark, XGBoost represents the most robust recommendation across both provisional and empirically derived weightings, and is a defensible default for resource-constrained financial institution deployment. However, the assumption that resource-constrained contexts require progressively larger accuracy concessions as budgets shrink is not supported by the case study evidence; practitioners' revealed preferences suggest a floor on acceptable detection performance that persists across budget tiers, with cost sensitivity operating above that floor rather than as a direct substitute for it.

5.4.4 Conclusion on RQ4: Decision Quality Improvement Over Accuracy-Only Evaluation

NIDS-EFS improves decision quality primarily by making previously invisible efficiency and cost trade-offs explicit and auditable, not by reliably shifting practitioners toward more efficiency-weighted outcomes than they would otherwise choose. This is a genuine but more modest contribution than a framework that changes the substantive decision, and it should be represented as such in any claims made about the framework's practical value.

5.5 Implications

5.5.1 Implications for Practice

- CISOs and IT procurement teams in African financial institutions have empirical, locally-relevant evidence that lightweight models such as XGBoost need not sacrifice detection performance for efficiency, directly supporting the cost-reduction case articulated in Chapter One.
- Total Cost of Ownership should be treated as a first-order procurement criterion rather than a secondary consideration; the near-doubling of its empirically elicited weight relative to the provisional assumption indicates practitioners already weight it heavily even before being shown a formal framework.
- Institutions should expect that presenting a transparent, efficiency-weighted recommendation will not automatically override stakeholders' preference for higher detection accuracy; procurement processes should budget for this preference rather than assume framework output alone will be decisive.

- Any ensemble or hybrid model under consideration should have its efficiency metrics independently verified before use in procurement decisions, given the data quality issue identified in this study's own ensemble benchmarking (Section 4.6.1).

5.5.2 Implications for Theory

- This study extends AHP's documented application domains (project selection, vendor evaluation, cybersecurity risk assessment; Saaty & Vargas, 2012) into technical ML model selection, contributing a worked example of AIJ aggregation compensating for low individual-level consistency in a domain-expert, non-AHP-trained panel.
- The divergence between elicited abstract criterion weights and revealed choice on concrete alternatives, observed here in a concrete applied setting, adds an empirical data point to the broader methodological caution that AHP practitioners should validate elicited weights against concrete choice scenarios rather than relying on pairwise comparisons alone.
- The four-dataset benchmarking design, and specifically the Naive Bayes anomaly it surfaced, provides methodological support for multi-dataset efficiency benchmarking as a validity safeguard against single-dataset overfitting of deployment recommendations.

5.5.3 Implications for Policy

- This study contributes what is, to the researcher's knowledge, among the first efficiency benchmark datasets for ML-based NIDS evaluated in an African financial institution context, providing national cybersecurity strategy makers with locally grounded evidence rather than extrapolation from resource-abundant contexts.
- The demonstrated feasibility of high-accuracy, low-resource NIDS models supports the equitable-access argument connecting this research to UN Sustainable Development Goals 9, 10, and 16, as articulated in Chapter One, by showing that advanced detection capability is achievable within realistic developing-economy budget constraints.

5.6 Limitations of the Study

Several limitations qualify the findings and conclusions above and should inform how they are used.

- The decision-support tool specified in RO3 has been partially delivered: a command-line interface (CLI) implementation is built and functional, accepting user-specified budget and deployment constraints and returning a ranked NIDS recommendation. Two elements of RO3 remain outstanding: the companion web application has not yet been built, and the formal System Usability Scale evaluation, targeting a score of at least 72 across twenty test participants, has not yet been conducted. The conclusions in this chapter concern the validity of the underlying NIDS-EFS scoring logic and its correct implementation in a working CLI tool, not the tool's usability or adoption-readiness in its final intended form.
- The Random Forest plus XGBoost ensemble's training time, inference latency, throughput, and memory measurements were recorded as exactly 0.0 across all four datasets, an unresolved benchmarking script gap rather than a genuine measurement (Section 4.6.1). Composite rankings involving this model, including its second-place position under both weighting schemes, should be treated with caution until re-benchmarked.
- The case study scenarios (Section 4.8) presented experts with illustrative, literature-derived performance and cost figures rather than the study's own empirically benchmarked values, a substitution flagged as provisional in the survey instrument but not corrected before administration. The case study therefore validates the framework's decision logic in principle rather than validating the specific empirical rankings produced in Section 4.7.
- The framework-implied scores used for comparison against expert case study choices (Section 4.8.3) could not incorporate Deployment Feasibility, the third-highest-weighted criterion, because the scenario text provided no measurable proxy for it; this comparison is therefore a partial rather than complete replication of the full five-criterion AHP scoring procedure.
- Only 43.8% of experts produced individually consistent pairwise comparison matrices; while the aggregated group matrix was highly consistent, the underlying individual judgments were, in most cases, not internally logical by Saaty's (1980) conventional standard, a limitation inherent to eliciting AHP judgments from busy practitioners rather than trained decision analysts.
- The expert panel, while exceeding the minimum recruitment target, was demographically concentrated: 68.8% reported more than ten years of experience, and the panel was weighted toward East African financial-institution practitioners. Findings may not generalize evenly to less senior practitioners or to other African regions and sectors.

- All benchmarking was conducted on a single CPU-only hardware configuration (Intel Core i7-1165G7, 16 GB RAM); no GPU-accelerated or edge-device (Raspberry Pi 4) validation, described in Chapter Three as an optional supplementary step, was performed.

5.7 Recommendations

5.7.1 Recommendations for Practice

- Financial institutions evaluating ML-based NIDS options should request efficiency metrics (latency, memory, energy, training time) alongside accuracy from vendors and internal development teams as standard procurement documentation, rather than accepting accuracy-only claims.
- Institutions with severe resource constraints should treat XGBoost, or comparable gradient-boosted tree ensembles, as a strong default starting point for evaluation, given its consistent top ranking across both weighting schemes examined in this study.
- Procurement teams should document both the framework-recommended alternative and the stakeholder-preferred alternative when they diverge, rather than treating the framework's output as automatically overriding practitioner judgment, consistent with this study's finding that transparency, not outcome substitution, is the framework's more defensible value proposition.

5.7.2 Recommendations for Further Research

- Build the companion web application for the NIDS-EFS decision-support tool and conduct the planned System Usability Scale evaluation with twenty participants on the existing CLI implementation, to close the remaining RO3 gap identified in Section 5.6.
- Re-run the ensemble benchmarking process to obtain genuine efficiency measurements for the Random Forest plus XGBoost ensemble, and recompute the composite AHP rankings once corrected data are available.
- Repeat the case study validation using the study's own empirically benchmarked performance and cost figures, rather than illustrative literature-derived values, and incorporate a measurable proxy for Deployment Feasibility into the scenario design.

- Conduct follow-up qualitative interviews with a subset of the expert panel to directly investigate the divergence between their elicited AHP weights and their concrete scenario choices, since the present design can document but not explain this divergence.
- Extend expert recruitment beyond the current panel's demographic concentration to include more junior practitioners and a wider range of African regions and organisational sectors, to test the generalizability of the empirically derived criterion weights.
- Validate the top-ranked models on GPU-accelerated and edge-device (Raspberry Pi 4) hardware, as originally scoped as an optional extension in Chapter Three, to extend the framework's applicability beyond the single CPU-only configuration tested here.

5.8 Chapter Summary

This chapter discussed the Chapter Four findings against the study's four research questions, concluding that computational efficiency and detection accuracy relate in an architecture- and dataset-dependent rather than uniform manner (RQ1); that AHP can be adapted into a standardized NIDS selection framework provided individual-level inconsistency is managed through group aggregation (RQ2); that XGBoost is a robust recommendation for resource-constrained deployment, though practitioners' revealed preferences do not simply scale detection accuracy down as budgets tighten (RQ3); and that the NIDS-EFS framework improves decision quality primarily through transparency and auditability rather than through reliably shifting practitioners toward more efficiency-weighted outcomes (RQ4). The chapter identified seven limitations, most significantly that the RO3 decision-support tool remains partially delivered, with a working CLI implementation but no web application and no formal usability evaluation yet completed, and that the Random Forest plus XGBoost ensemble's efficiency data requires correction. Six recommendations for practice and further research were offered, prioritizing completion of the web application and System Usability Scale evaluation, correction of the ensemble benchmarking gap, and a repeat case study validation using the study's own empirical figures. Together, these findings and recommendations substantiate the study's central contribution: a validated, empirically weighted, and honestly limited framework for closing the accuracy-feasibility divide in NIDS selection for resource-constrained financial institutions.

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APPENDIX

APPENDIX A: PARTICIPANT INFORMATION SHEET

Study Title	A Multi-Criteria Decision Framework for NIDS Selection in Resource-Constrained Environments
Researcher	Shema Prince | ID: 663029
Program	MSc Information Security - ISR 6890
Institution	USIU-Africa, School of Science and Technology
Contact	pshema@usiu.ac.ke / Shemaprince123@gmail.com

What is this study about?

This research develops the NIDS Efficiency-First Selection (NIDS-EFS) framework, a multi-criteria decision tool that helps cybersecurity practitioners in resource-constrained African financial institutions select suitable Network Intrusion Detection Systems. You are invited because of your professional experience in network security, information security governance, cybersecurity consulting, security research, or IT security management.

What will I be asked to do?

You will complete a short online survey (25–35 minutes) with three sections:

- Section A: Professional demographics (role, years of experience, sector, region, no name or employer required).

- Section B: 10 AHP pairwise comparison questions rating the importance of five NIDS evaluation criteria using Saaty's 1–9 scale. Criteria are clearly defined with practical examples.
- Section C: Two short deployment scenarios asking which NIDS option you would recommend.

Is participation voluntary?

Yes. Participation is entirely voluntary. You may withdraw at any time without consequence. If you wish to withdraw after submitting, contact the researcher within 14 days and your data will be excluded.

How will my data be protected?

- No names, employer details, or contact information will be retained beyond consent.
- Responses are anonymous and stored in encrypted, password-protected storage.
- Results will be reported in aggregate only, no individual response will be identifiable.
- Data will be retained for 5 years after thesis submission then permanently deleted, in line with USIU-Africa policy and the Kenya Data Protection Act (2019).

Ethical Approval

This study has been approved by the USIU-Africa Institutional Review Board (IRB). For concerns about the ethical conduct of this research, contact the USIU-Africa Office of Research and Innovation.

Thank you for considering participation in this research.

APPENDIX B: INFORMED CONSENT FORM

Study Title	A Multi-Criteria Decision Framework for NIDS Selection in Resource-Constrained Environments
Researcher	Shema Prince | ID: 663029 | ISR 6890 | USIU-Africa

Please read each statement and tick the box to confirm your understanding and agreement before signing.

☐ I have read and understood the Participant Information Sheet (Appendix B1) and had the opportunity to ask questions.

☐ I understand that participation is voluntary and I may withdraw at any time without consequence.

☐ I understand that no personally identifiable information will be collected or retained beyond the consent stage.

☐ I understand my anonymised data will be retained securely for 5 years then permanently deleted, per USIU-Africa policy and the Kenya Data Protection Act (2019).

☐ I understand results will be reported in aggregate only and my individual responses will never be identifiable.

☐ I confirm I have relevant professional experience in network security, information security, cybersecurity, or IT security management.

☐ I agree to participate in this study and complete the AHP survey.

PARTICIPANT

Signature: _____ Date: _____

Full Name (optional, may be left blank): _____

RESEARCHER

Signature: _____ Date: _____

Name: Shema Prince | ID: 663029 | pshema@usiu.ac.ke / shemaprince123@gmail.com

Please retain a copy of this form and the Participant Information Sheet for your records.

APPENDIX C: OFFICIAL DEBRIEF FORM

Thank you for participating in the USIU-Africa study titled **"A Multi-Criteria Decision Framework for Network Intrusion Detection System Selection in Resource-Constrained Environments."**

This research is being conducted by **Shema Prince**, a Master of Science in Information Security student at United States International University – Africa (Student ID: 663029; Course: ISR 6892), as part of the requirements for the MSc Information Security degree.

The purpose of this study is to develop and validate the **NIDS Efficiency-First Selection (NIDS-EFS) Framework**, a multi-criteria decision analysis tool designed to support evidence-based Network Intrusion Detection System (NIDS) selection within resource-constrained African financial institutions.

Your participation involved providing expert judgment through an AHP pairwise comparison survey. All responses remain completely anonymous and confidential. No personally identifiable information, including names, employer details, or contact information, has been retained beyond the consent stage. Data has been recorded and stored only in aggregated form for academic analysis.

All collected data is securely stored in encrypted, password-protected storage accessible only to the principal researcher. In accordance with the Kenya Data Protection Act (2019) and USIU-Africa research policy, data will be retained for five (5) years after thesis submission and permanently deleted thereafter.

This study has been reviewed and approved by the USIU-Africa Institutional Review Board (IRB). Participation in this research is voluntary, and participants may withdraw their consent within fourteen (14) days of survey completion by contacting the researcher. Upon withdrawal, submitted responses will be excluded from the analysis.

If you have any questions regarding this research or wish to request a summary of the findings, please contact the researcher or the USIU-Africa IRB office using the details below.

Researcher Shema Prince

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APPENDIX D: AHP EXPERT SURVEY QUESTIONNAIRE

NIDS Efficiency-First Selection (NIDS-EFS) Framework

Multi-Criteria Decision Analysis - AHP Pairwise Comparison Survey

Study Title: A Multi-Criteria Decision Framework for NIDS Selection in Resource-Constrained Environments

Researcher: Shema Prince | ID: 663029 | MSc Information Security | USIU-Africa

Contact: pshema@usiu.ac.ke / shemaprince123@gmail.com

Ethics Approval: USIU-Africa IRB

This survey is part of a research study developing the NIDS Efficiency-First Selection (NIDS-EFS) framework. The framework uses the Analytic Hierarchy Process (AHP) to help cybersecurity practitioners in resource-constrained African financial institutions make evidence-based Network Intrusion Detection System deployment decisions. Your professional judgment is essential to establishing the relative importance of the five evaluation criteria.

This survey takes approximately 25–35 minutes to complete. All responses are anonymous and confidential. Please read each question carefully before responding.

SECTION A: Professional Demographics

This section collects background information to describe the participant panel. No personally identifiable information is requested. All fields are optional but help ensure panel diversity.

A1. Years of professional experience in cybersecurity or IT security	☐ Less than 3 years ☐ 3–5 years ☐ 6–10 years

	☐ More than 10 years
A2. Current functional role	☐ Network / Security Analyst ☐ IT Security Manager / CISO ☐ Cybersecurity Consultant ☐ Academic / Researcher ☐ Other: _____
A3. Organisational sector	☐ Financial institution (bank, microfinance, fintech) ☐ Telecommunications ☐ Government / Public sector ☐ Academia / Research institution ☐ Other: _____
A4. Geographic region	☐ East Africa (Kenya, Uganda, Tanzania, Rwanda, Ethiopia) ☐ West Africa ☐ Southern Africa ☐ Other African region ☐ Outside Africa
A5. Organisational size by annual cybersecurity budget	☐ Small (under $10,000 / year) ☐ Mid-size ($10,000–$50,000 / year) ☐ Large (above $50,000 / year)

SECTION B: Criteria Definitions

Before completing the pairwise comparison questions in Section C, please review the five NIDS-EFS evaluation criteria defined below. These criteria represent the dimensions on which ML-based NIDS models will be scored and ranked by the framework.

Code	Criterion	Definition and Operational Example
C1	**Detection Performance**	How accurately the NIDS detects attacks, including minority classes. Measured by F1-score, per-class precision and recall across all attack categories. Example: a model with 98% overall accuracy but 0% recall on U2R attacks scores poorly on this criterion.
C2	**Computational Efficiency**	How quickly the NIDS classifies network traffic in real time. Measured by inference latency (milliseconds per sample) and throughput (samples per second). Example: a model taking 500ms per classification cannot support real-time detection on high-traffic networks.
C3	**Deployment Feasibility**	Whether the NIDS can be installed and operated on the organisation's existing hardware and staff expertise. Measured by hardware compatibility (CPU-only vs GPU required), configuration complexity, and maintenance burden. Example: a deep learning model requiring specialised ML engineers to maintain scores poorly if the organisation has only general IT staff.
C4	**Resource Consumption**	The memory, storage, and energy the NIDS consumes during operation. Measured by RAM footprint (GB), model size (MB), and energy draw (Watts). Example: a model requiring 12GB RAM cannot run on a shared server with 16GB total RAM that also runs other services.
C5	**Total Cost of Ownership**	The full lifecycle cost of deploying and maintaining the NIDS over 3–5 years. Includes hardware costs, energy bills, retraining time, software updates, and staff training. Example: a model requiring

		quarterly retraining at 8 hours per cycle has a substantially higher TCO than one requiring annual updates.

SECTION C: AHP Pairwise Comparison Questions

Instructions: For each pair of criteria below, circle or tick the number that best reflects your professional judgment about their relative importance for NIDS selection in a resource-constrained African financial institution. Use the Saaty 1–9 scale shown below.

Scale Value	Verbal Judgement	Definition
1	Equal Importance	Two criteria contribute equally to the objective
3	Moderate Importance	Experience slightly favours one criterion over another
5	Strong Importance	Experience strongly favours one criterion over another
7	Very Strong Importance	One criterion is strongly favoured and dominance is demonstrated
9	Extreme Importance	The evidence favouring one criterion over another is of the highest possible order
2, 4, 6, 8	Intermediate Values	Used when compromise between adjacent judgements is needed

Note: Numbers to the LEFT of centre (1) indicate Criterion A is more important. Numbers to the RIGHT indicate Criterion B is more important. The centre value 1 indicates equal importance.

Q1	Comparing: **Detection Performance (C1)** vs **Computational Efficiency (C2)**																
← Favours Criterion A	Favours Criterion B →																
A: Detection Performance (C1) B: Computational Efficiency (C2)	9	8	7	6	5	4	3	2	**1**	2	3	4	5	6	7	8	9

Q2	Comparing: **Detection Performance (C1)** vs **Deployment Feasibility (C3)**																
← Favours Criterion A	Favours Criterion B →																
A: Detection Performance (C1) B: Deployment Feasibility (C3)	9	8	7	6	5	4	3	2	**1**	2	3	4	5	6	7	8	9

Q3	Comparing: **Detection Performance (C1)** vs **Resource Consumption (C4)**																
← Favours Criterion A	Favours Criterion B →																
A: Detection Performance (C1) B: Resource Consumption (C4)	9	8	7	6	5	4	3	2	**1**	2	3	4	5	6	7	8	9

Q4	Comparing: **Detection Performance (C1)** vs **Total Cost of Ownership (C5)**																
← Favours Criterion A	Favours Criterion B →																
A: Detection Performance (C1) B: Total Cost of Ownership (C5)	9	8	7	6	5	4	3	2	**1**	2	3	4	5	6	7	8	9

Q5	Comparing: **Computational Efficiency (C2)** vs **Deployment Feasibility (C3)**																
← Favours Criterion A	Favours Criterion B →																
A: Computational Efficiency (C2) B: Deployment Feasibility (C3)	9	8	7	6	5	4	3	2	**1**	2	3	4	5	6	7	8	9

Q6	Comparing: **Computational Efficiency (C2)** vs **Resource Consumption (C4)**																
← Favours Criterion A	Favours Criterion B →																
A: Computational Efficiency (C2)	9	8	7	6	5	4	3	2	**1**	2	3	4	5	6	7	8	9

B: Resource Consumption (C4)																	

Q7	Comparing: **Computational Efficiency (C2)** vs **Total Cost of Ownership (C5)**																
← Favours Criterion A	Favours Criterion B →																
A: Computational Efficiency (C2)	9	8	7	6	5	4	3	2	**1**	2	3	4	5	6	7	8	9
B: Total Cost of Ownership (C5)																	

Q8	Comparing: **Deployment Feasibility (C3)** vs **Resource Consumption (C4)**																
← Favours Criterion A	Favours Criterion B →																
A: Deployment Feasibility (C3)	9	8	7	6	5	4	3	2	**1**	2	3	4	5	6	7	8	9
B: Resource Consumption (C4)																	

Q9	Comparing: **Deployment Feasibility (C3)** vs **Total Cost of Ownership (C5)**																
← Favours Criterion A	Favours Criterion B →																
A: Deployment Feasibility (C3)	9	8	7	6	5	4	3	2	**1**	2	3	4	5	6	7	8	9
B: Total Cost of Ownership (C5)																	

Q10	Comparing: **Resource Consumption (C4)** vs **Total Cost of Ownership (C5)**																
← Favours Criterion A	Favours Criterion B →																
A: Resource Consumption (C4)	9	8	7	6	5	4	3	2	**1**	2	3	4	5	6	7	8	9
B: Total Cost of Ownership (C5)																	

SECTION D: Deployment Scenario Validation

The following two scenarios present hypothetical NIDS deployment situations. Based on your professional experience, indicate which NIDS option you would recommend for each scenario. This section validates the framework's output against practitioner judgment.

Scenario 1 — Small Fintech Startup

Organisation: Digital payments fintech, East Africa

Annual security budget: Under $10,000

Hardware: Single CPU-only laptop server, 8GB RAM

Staff: One generalist IT officer, no ML expertise

Traffic volume: Low — under 10,000 network flows per day

Which model would you recommend?

☐ Option A: Random Forest + SMOTE (90% F1-score, 4ms latency, 45MB RAM, $800/year TCO)

☐ Option B: Decision Tree (85% F1-score, 1ms latency, 12MB RAM, $400/year TCO)

☐ Option C: CNN + ADASYN (98% F1-score, 250ms latency, 4.2GB RAM, $6,500/year TCO)

Reason for your choice (optional):

_____

Scenario 2 — Mid-Size Commercial Bank

Organisation: Licensed commercial bank, multiple branches

Annual security budget: $20,000–$50,000

Hardware: Entry-level server, CPU-only, 32GB RAM

Staff: Small IT security team, limited ML exposure

Traffic volume: Medium — 10,000–100,000 network flows per day

Which model would you recommend?

☐ Option A: XGBoost + ADASYN (94% F1-score, 22ms latency, 380MB RAM, $1,200/year TCO)

☐ Option B: Ensemble RF+XGB (97% F1-score, 48ms latency, 820MB RAM, $2,800/year TCO)

☐ Option C: LSTM + SMOTE (96% F1-score, 380ms latency, 3.8GB RAM, $7,200/year TCO)

Reason for your choice (optional):

_____

***Note**: Performance metrics shown in the scenarios are illustrative estimates based on published literature and will be replaced with empirically measured values from the NIDS-EFS benchmarking phase prior to the actual survey administration.*

SECTION E: Additional Comments

Do you have any comments on the five criteria, their definitions, or the framework concept? (Optional)

Thank you for completing this survey. Your expert judgment directly contributes to validating the NIDS-EFS framework and advancing evidence-based cybersecurity practice in resource-constrained African financial institutions.

To request a summary of the research findings upon completion, please contact: pshema@usiu.ac.ke / Shemaprince123@gmail.com

APPENDIX E: PROJECT PLAN AND RESEARCH BUDGET

Student: Shema | ID: 663029 | Course: ISR 6890 | Supervisor: Stanley M. Githinji

Project Title: A Framework for Efficiency-First Network Intrusion Detection System (NIDS) Selection in Resource-Constrained African Financial Institutions

1. Project Overview

This appendix presents my project plan and research budget for the NIDS Efficiency-First Selection (NIDS-EFS) framework study. My research runs across three semesters: Semester 1 for the proposal stage (14 weeks), Semester 2 for experimentation and field work (14 weeks), and Semester 3 for writing and the final defense (14 weeks). The budget covers all direct costs I expect to incur across data collection, software, expert survey administration, computing, and dissemination. All costs are in Kenya Shillings (KES).

2. Work Breakdown Structure (WBS)

I have broken my research down into five work packages that match my objectives and my three-semester plan.

Table A1: Work Packages and Deliverables

WP	Work Package	Key Activities and Deliverables	Semester	Duration
WP1	Literature Review and Proposal	Chapter 1 (Introduction); Chapter 2 (Literature Review); Chapter 3 (Methodology); proposal defense slides and presentation	Semester 1	14 weeks
WP2	IRB and Dataset Preparation	IRB application and clearance; expert survey instrument development and pilot testing; dataset download and SHA-256 verification; preprocessing pipeline setup and testing	Semester 2, Weeks 1 to 4	4 weeks
WP3	Benchmarking Simulations	Execute 24 model-dataset experiments (8 models by 3 datasets); measure 8 metrics per experiment (192 total measurements); SMOTE and ADASYN overhead measurement; MLflow experiment tracking	Semester 2, Weeks 3 to 9	7 weeks
WP4	Expert Survey and AHP Analysis	Recruit 15 to 20 cybersecurity practitioners; administer AHP pairwise comparison survey; aggregate weights using geometric mean; consistency ratio verification; sensitivity analysis; Monte Carlo simulation; scenario validation	Semester 2, Weeks 3 to 12	10 weeks
WP5	Writing, Review and Defense	Chapter 4 (Results); Chapter 5 (Discussion); Chapter 6 (Conclusions and Recommendations); abstract; full document compilation; supervisor review cycles; submission to PD; final oral defense	Semester 3	14 weeks

3. Research Budget

I have organised my budget into five cost categories following the ISR6890 project planning guidelines: personnel, equipment and software, data collection, travel and logistics, and miscellaneous costs. Each line is linked to the work package it supports.

3.1 Personnel Costs

I am conducting this research on my own without any external research assistants. The consultation cost is included in case I need targeted statistical guidance on the AHP methodology from a faculty member during the analysis phase.

Table A2: Personnel Costs

Item	Unit	Qty	Unit Cost (KES)	Total (KES)
Principal Researcher (self-funded)	Semester	3	0	0
Statistical or Methodological Consultation (if needed)	Session	2	5,000	10,000
Language Proofreading (final document, as required by USIU handbook)	Document	1	8,000	8,000
Thesis Hard Binding (3 copies required for submission)	Copy	3	1,000	3,000
Personnel Subtotal				**KES 21,000**

3.2 Equipment and Software Costs

I will run all benchmarking simulations on my personal laptop, an Intel Core i7-1165G7 at 2.80 GHz with 16 GB RAM running Windows 11 Pro, which I purchased for this and other academic work. I have included the laptop cost in Table A3 to give a complete picture of the total equipment investment in this project. No additional hardware purchases are needed.

Table A3: Equipment and Software Costs

Item	Unit	Qty	Unit Cost (KES)	Total (KES)
Personal Laptop (Intel Core i7-1165G7, 16 GB RAM, Windows 11 Pro, purchased for academic and research use)	Unit	1	95,000	95,000
Python 3.10 plus scikit-learn, TensorFlow, MLflow (all open source, no cost)	N/A	N/A	0	0
Benchmark datasets: NSL-KDD, CICIDS2017, UNSW-NB15 (publicly available, no cost)	N/A	N/A	0	0
Microsoft Office 365 (Provided by USIU-Africa)	N/A	N/A	0	0
Cloud backup storage (Google Drive or OneDrive)	Year	1	2,500	2,500
External hard drive for data backup (1 TB)	Unit	1	5,500	5,500
Electricity and internet costs (incremental, running simulations over 10 months)	Month	10	2,000	20,000
Equipment and Software Subtotal				**KES 123,000**

3.3 Data Collection Expenses

These costs cover running my expert survey and reaching the 15 to 20 qualified cybersecurity practitioners I need. I plan to use Google Forms which is free, and offer each participant a small token of appreciation for their time.

Table A4: Data Collection Expenses

Item	Unit	Qty	Unit Cost (KES)	Total (KES)
Online survey platform (Google Forms, free tier)	N/A	N/A	0	0
Participant appreciation token (20 experts at KES 1,000 each)	Participant	20	1,000	20,000

Printed survey copies (physical backup)	Copy	25	50	1,250
Follow-up interview communication (calls and mobile data)	Lump sum	1	2,000	2,000
Benchmark datasets (open access, no cost)	N/A	N/A	0	0
Data Collection Subtotal				**KES 23,250**

3.4 Travel and Logistics

I have budgeted for transport to and from campus for supervisor meetings, trips to recruit participants through ISACA Kenya and other professional networks, and both defense days. I have also included an optional conference cost in case I get the opportunity to present my findings.

Table A5: Travel and Logistics Costs

Item	Unit	Qty	Unit Cost (KES)	Total (KES)
Supervisor meetings (transport to and from campus, estimated 20 trips)	Trip	20	400	8,000
Participant recruitment visits (ISACA Kenya and industry contacts)	Trip	5	1,000	5,000
Proposal defense day (transport and refreshments)	Day	1	2,000	2,000
Final defense day (transport and refreshments)	Day	1	2,000	2,000
Conference or seminar attendance (optional, for disseminating findings)	Event	1	10,000	10,000
Travel and Logistics Subtotal				**KES 27,000**

3.5 Miscellaneous and Contingency Costs

This covers printing costs for drafts and the final thesis, any IRB-related admin fees, research consumables like binders, stationery, and printing supplies, and a 10% contingency that I set aside in case something comes up that I have not planned for.

Table A6: Miscellaneous and Contingency Costs

Item	Unit	Qty	Unit Cost (KES)	Total (KES)
Printing and photocopying (draft chapters and final thesis copies)	Lump sum	1	5,000	5,000
IRB application administrative fee (if applicable)	Lump sum	1	2,000	2,000
Research consumables (binders, stationery, pens, printing supplies)	Lump sum	1	2,000	2,000
Contingency reserve (10% of total direct costs: KES 203,250)	Lump sum	1	20,325	20,325
Miscellaneous and Contingency Subtotal				**KES 29,325**

3.6 Total Budget Summary

Table A7: Total Research Budget Summary

Cost Category	Work Packages	Total (KES)	%
Personnel Costs	WP1, WP5	KES 21,000	9.4%
Equipment and Software	WP1 to WP5	KES 123,000	55.0%
Data Collection Expenses	WP2, WP4	KES 23,250	10.4%
Travel and Logistics	WP1 to WP5	KES 27,000	12.1%
Miscellaneous and Contingency	WP1 to WP5	KES 29,325	13.1%
TOTAL PROJECT BUDGET		**KES 223,575**	**100%**

Note: Most technical costs are zero because all the tools and datasets I am using are open source and freely available. The main cost drivers are my laptop (which I purchased for this work), participant incentives for the expert survey, electricity for overnight simulation runs, and the 10% contingency buffer.

4. Budget Justification

Here is my reasoning for each cost in the budget:

1. Language proofreading (KES 8,000): The USIU-Africa Research Handbook requires every final thesis to go through proofreading by a language expert before submission, so this is not optional for me.

2. Thesis hard binding (KES 3,000): The handbook requires me to submit three hard-bound copies to the Dean through the Programme Director. At KES 1,000 per copy, that is a reasonable rate for academic binding in Nairobi.

3. Statistical consultation (KES 10,000): I have included two sessions in case I need expert guidance on the AHP eigenvalue computation or any part of the statistical analysis during Semester 2.

4. Personal laptop (KES 95,000): I purchased my Intel Core i7 laptop specifically to handle the computational demands of this research alongside my studies. I included it here to give a complete and honest picture of the total investment going into this project.

5. Microsoft Office 365 (KES 0): USIU-Africa provides Microsoft 365 to all enrolled students at no personal cost. I access it through my student login and use it for thesis writing, table formatting, and figure preparation.

6. Cloud backup and external hard drive (KES 8,000): The CICIDS2017 dataset alone is around 500 MB, and I will be generating a lot of simulation output files. Having both cloud and a physical backup drive protects me against losing months of work.

7. Electricity and internet (KES 20,000): Running CNN and LSTM training overnight on large datasets can take 6 to 8 hours per session over roughly 10 months. That adds up, so I have budgeted for the extra electricity and data costs that come with it.
8. Participant appreciation tokens (KES 20,000): To get 15 to 20 qualified experts to give up 25 to 35 minutes of their time completing my AHP survey, a small appreciation token of KES 1,000 per participant is a reasonable and fair amount.
9. Travel (KES 27,000): This covers transport for supervisor meetings throughout three semesters, recruitment trips, both defense days, and an optional conference in case I get the chance to present my findings.
10. Contingency reserve (KES 20,325): I set aside 10% of my total direct costs as a buffer for anything I have not anticipated, whether that is needing extra recruitment efforts, a software cost going up, or a minor equipment issue.

5. Risk Register

Below are the key risks I have identified for this project, along with what I plan to do if each one happens.

Table A8: Risk Register

ID	Risk	Likelihood	Impact	My Mitigation Plan
R1	Expert survey recruitment falls below 15 participants	Medium	High	I will start building my participant list now and begin outreach 4 weeks before the survey opens. I will use snowball sampling through ISACA Kenya. If needed, I can justify a minimum of 10 per Saaty and Vargas (2012).

R2	IRB approval delayed beyond Week 2 of Semester 2	Medium	High	I will submit my IRB application during Semester 1 Week 13, right after the proposal defense, with all consent forms already prepared so there is no waiting around.
R3	CNN and LSTM training exceeds my laptop capacity	Medium	Medium	I will run neural models overnight. If CICIDS2017 is too large, I will use a stratified sample and present the deep learning results as preliminary findings.
R4	Benchmark dataset download or integrity failure	Low	Medium	I will verify every download with SHA-256 hashing, keep two backup copies in separate locations, and store the original checksums.
R5	Laptop hardware failure during simulations	Low	High	I will commit to Git every week and keep MLflow logs on cloud backup. The university computer lab is my fallback if my laptop goes down.
R6	Supervisor feedback takes longer than 10 working days	Low	Medium	I will agree on a feedback schedule with my supervisor at the start of each semester and submit chapter drafts early. If delays go beyond the handbook limit, I will escalate to the Programme Director.
R7	AHP consistency ratio exceeds 0.10 in expert responses	Medium	Low	I will return any problematic matrices to the experts with specific guidance on which pairs are inconsistent. My sensitivity analysis also accounts for weight uncertainty across all valid responses.
R8	Budget overrun due to unexpected costs	Low	Low	My 10% contingency covers this. If it gets tight, I can defer the optional conference attendance.

6. Summary

This appendix covers my full project plan and budget for the NIDS-EFS framework study. My total projected budget is KES 223,575 spread across five cost categories. Most of the technical costs are zero because all the tools and datasets I am using are open source and freely available. I have broken the work into five packages tied to my three-semester timeline, identified eight risks with clear plans to handle each one, and included a Gantt chart on the next page showing exactly when I plan to do everything.

Table A9: Project Gantt Chart — Three-Semester Timeline

M1 to M4: Semester 1 (Proposal) | M5 to M8: Semester 2 (Experimentation) | M9 to M12: Semester 3 (Writing and Defense)

Work Package / Activity	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
SEMESTER 1: PROPOSAL STAGE												
Chapter 1: Introduction												
Chapter 2: Literature Review												
Chapter 3: Methodology												
Proposal Defense Preparation												
Proposal Defense												
SEMESTER 2: EXPERIMENTATION AND FIELD WORK												
IRB Application and Clearance												
Dataset Download, Verification and Preprocessing												
Expert Survey Instrument Development												
Expert Recruitment and Survey Administration												
Benchmarking Simulations (Traditional ML Models)												
Benchmarking Simulations (Deep Learning Models)												
AHP Weight Analysis and Consistency Check												
Sensitivity Analysis and Scenario Validation												
Chapter 4 Draft (Results)												
SEMESTER 3: WRITING AND FINAL DEFENSE												
Finalise Chapter 4 (Supervisor Review)												

Chapter 5: Discussion												
Chapter 6: Conclusions and Recommendations												
Abstract, Front Matter and Full Document Compilation												
Supervisor Final Review and Corrections												
Submission to Programme Director (2 weeks before defense)												
Final Oral Defense												

Shaded cells show when each activity is active. Semester 1 tasks are in dark blue, Semester 2 in green, and Semester 3 in navy.